

Data Exploration and Visualization

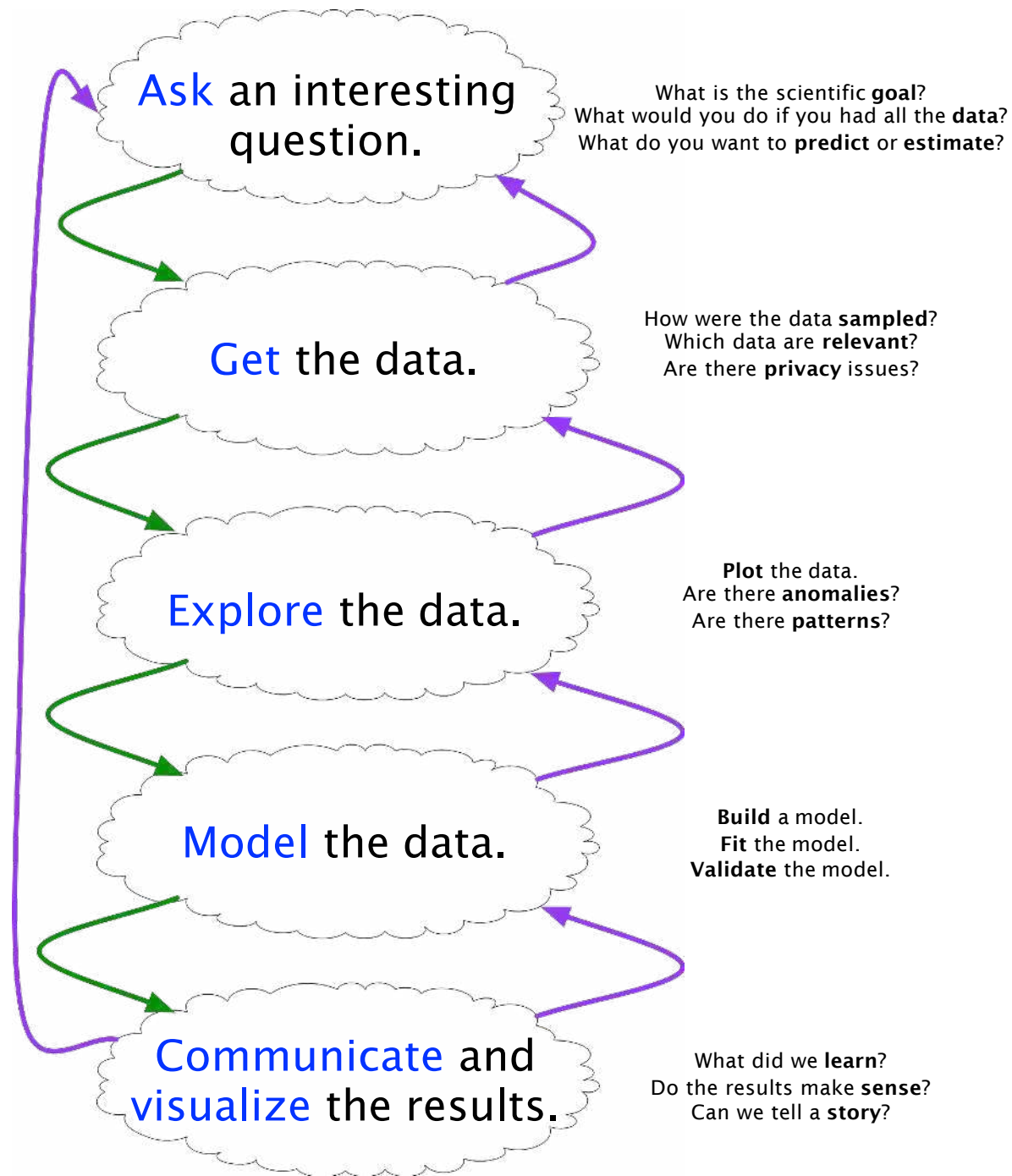
Data Exploration

Not always sure what we are looking for
(until we find it)



Visualization

- Visualization is the conversion of data into a visual or tabular format.
- Visualization helps understand the characteristics of the data and the relationships among data items or attributes can be analyzed or reported.
- Visualization of data is one of the most powerful and appealing techniques for data exploration.
 - Humans have a well developed ability to analyze large amounts of information that is presented visually
 - Can detect general patterns and trends
 - Can detect outliers and unusual patterns

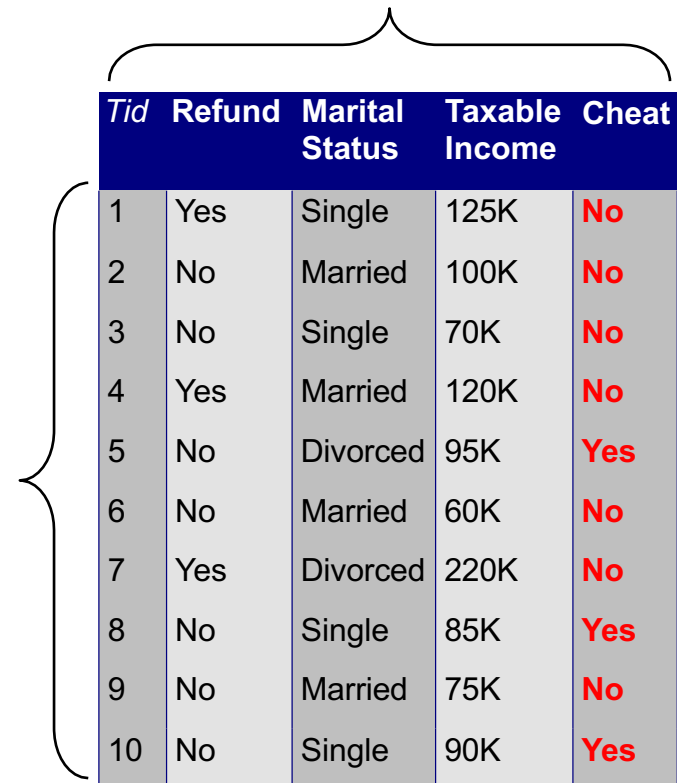


What is Data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
 - Examples: eye color of a person, temperature, etc.
 - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
 - Object is also known as record, point, case, sample, entity, or instance

Objects

Attributes



<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Type of variables (attributes)

- **Descriptive (categorical) variables**

- **Nominal** variables (no order between values): gender, eye color, race group, ...
- **Ordinal** variables (inherent order among values): response to treatment: none, slow, moderate, fast

- **Measurement variables**

- **Continuous measurement** variable: height, weight, blood pressure ...
- **Discrete measurement** variable (values are integers): number of siblings, the number of times a person has been admitted to a hospital ...

Properties of Attribute Values

- The type of an attribute depends on which of the following properties it possesses:
 - Distinctness: $= \neq$
 - Order: $< >$
 - Addition: $+ -$
 - Multiplication: $* /$
 - Nominal attribute: distinctness
 - Ordinal attribute: distinctness & order
 - Interval attribute: distinctness, order & addition
 - Ratio attribute: all 4 properties

The Trouble with Summary Stats

Set A		Set B		Set C		Set D	
X	Y	X	Y	X	Y	X	Y
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.11	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

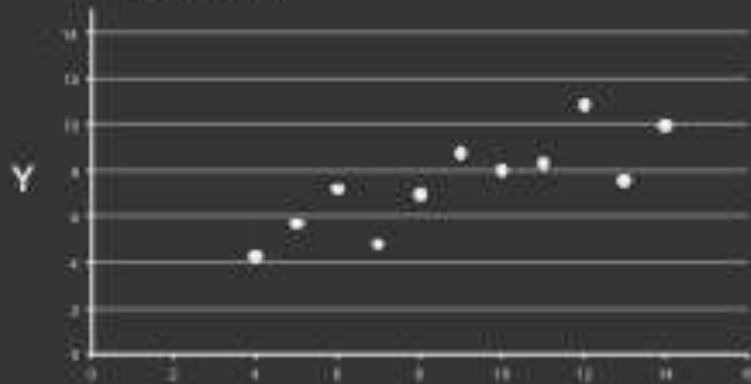
Summary Statistics Linear Regression

$$\begin{aligned}
 u_X &= 9.0 & \sigma_X &= 3.317 & Y &= 3 + 0.5 X \\
 u_Y &= 7.5 & \sigma_Y &= 2.03 & R^2 &= 0.67
 \end{aligned}$$

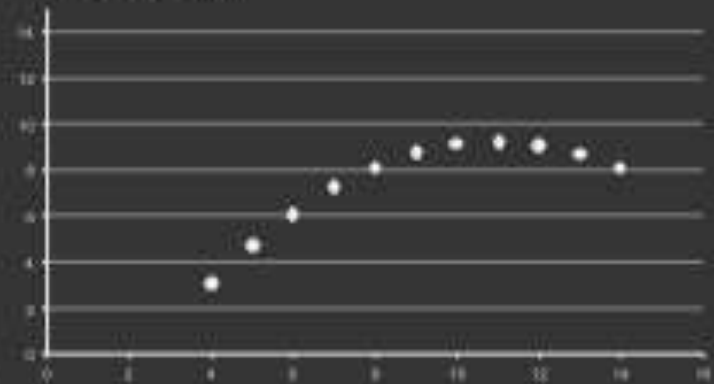
[Anscombe 73]

Looking at Data

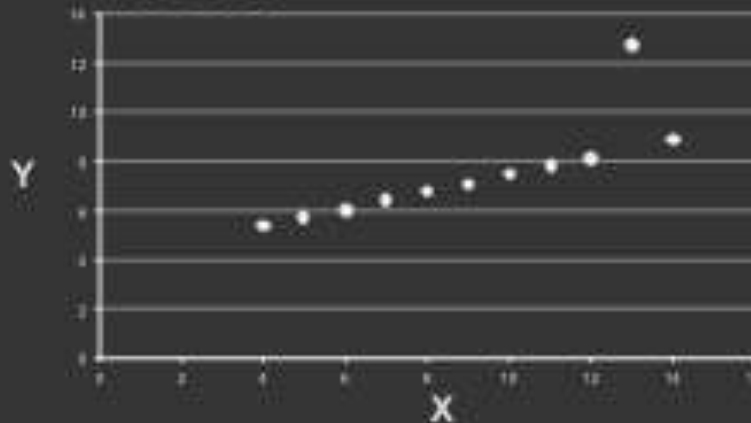
Set A



Set B



Set C



Set D

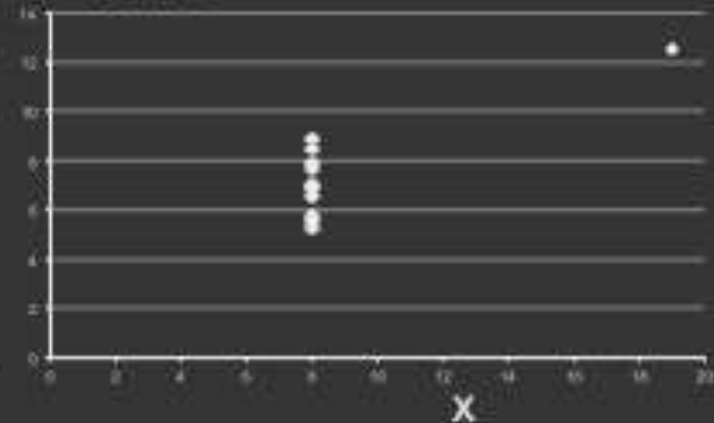


Chart types

- Single variable
 - Dot plot
 - Box-and-whisker plot
 - Histogram
 - Jitter plot
 - Error bar plot
 - Cumulative distribution function

Chart types

- Two variables
 - Bar chart
 - Scatter plot
 - Line plot
 - Log-log plot
- More than two variables
 - Stacked plots
 - Parallel coordinate plot

Sample Data

Height	Weight	Waist	Hip	bp.sys	bp.dia
172	72	87	94	127.5	80
166	91	109	107	172.5	100
174	80	95	101	123	64
176	79	93	100	117	76
166	55	70	94	100	60
163	76	96	99	160	87.5
154	84	98	118	130	80
165	90	108	101	139	80
155	66	80	96	120	70
146	59	77	96	112.5	75
164	62	76	93	130	47.5
159	59	76	96	109	69
163	69	96	99	155	100
143	73	97	117	137.5	85
. . .					

Plotting a Vector

- *plot(v)* will print the elements of the vector *v* according to their index

Plot height for each observation

> plot(dataset\$Height)

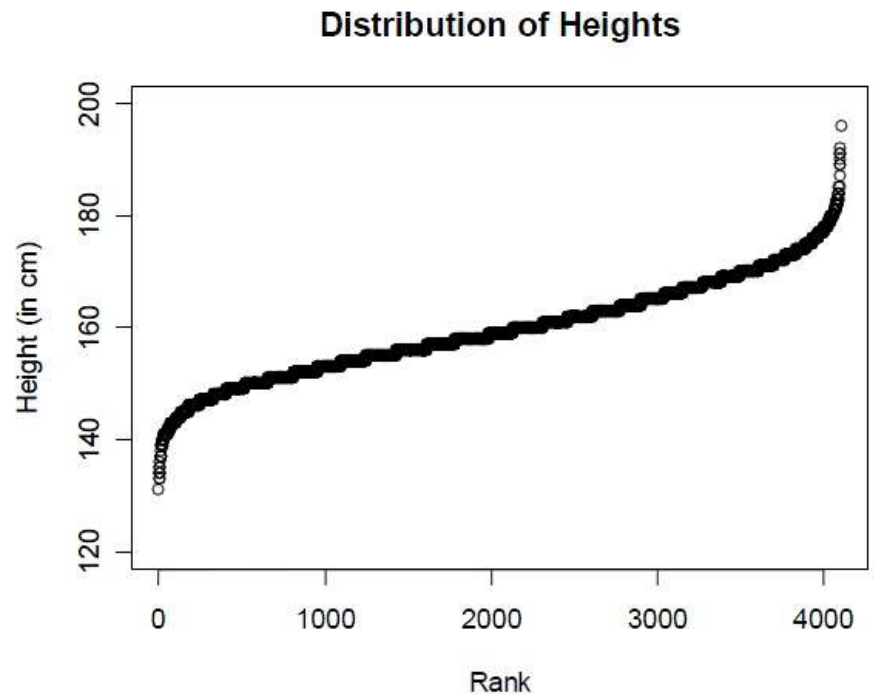
Plot values against their ranks

> plot(sort(dataset\$Height))

Parameters for plot()

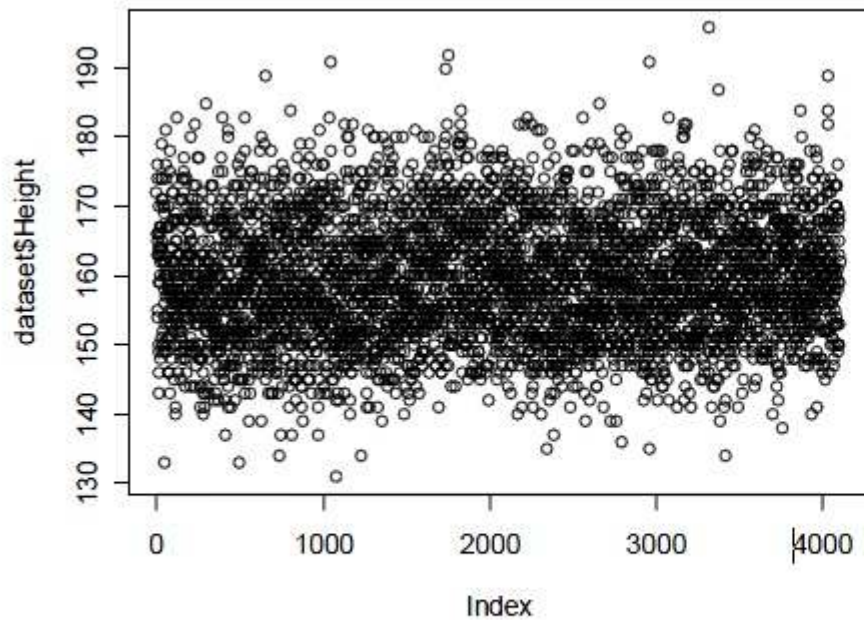
- Specifying labels:
 - main – provides a title
 - xlab – label for the x axis
 - ylab – label for the y axis

- Specifying range limits
 - ylim – 2-element vector gives range for y axis
 - xlim – 2-element vector gives range for x axis

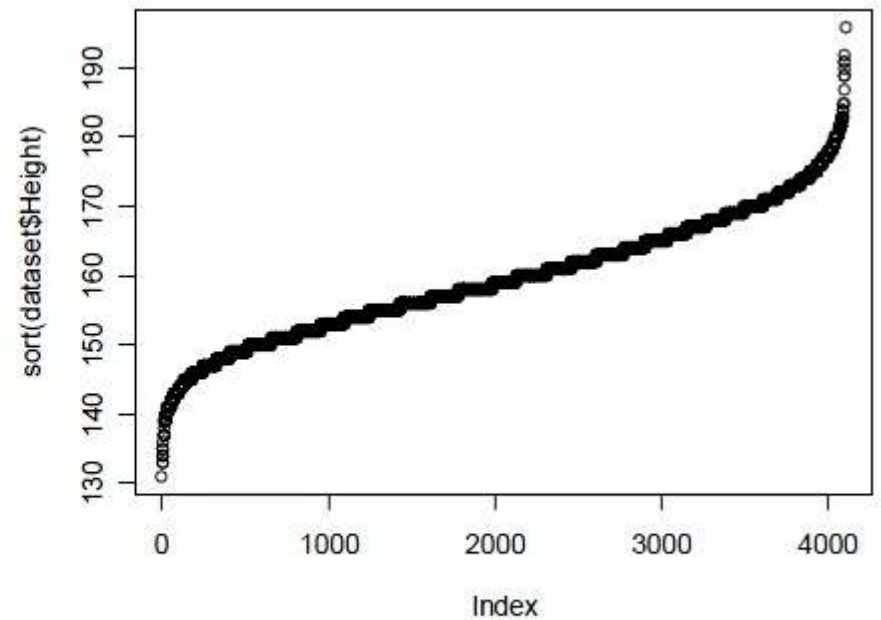


```
plot(sort(dataset$Height), ylim = c(120,200),  
     ylab = "Height (in cm)", xlab = "Rank", main = "Distribution of Heights")
```

Plotting a Vector



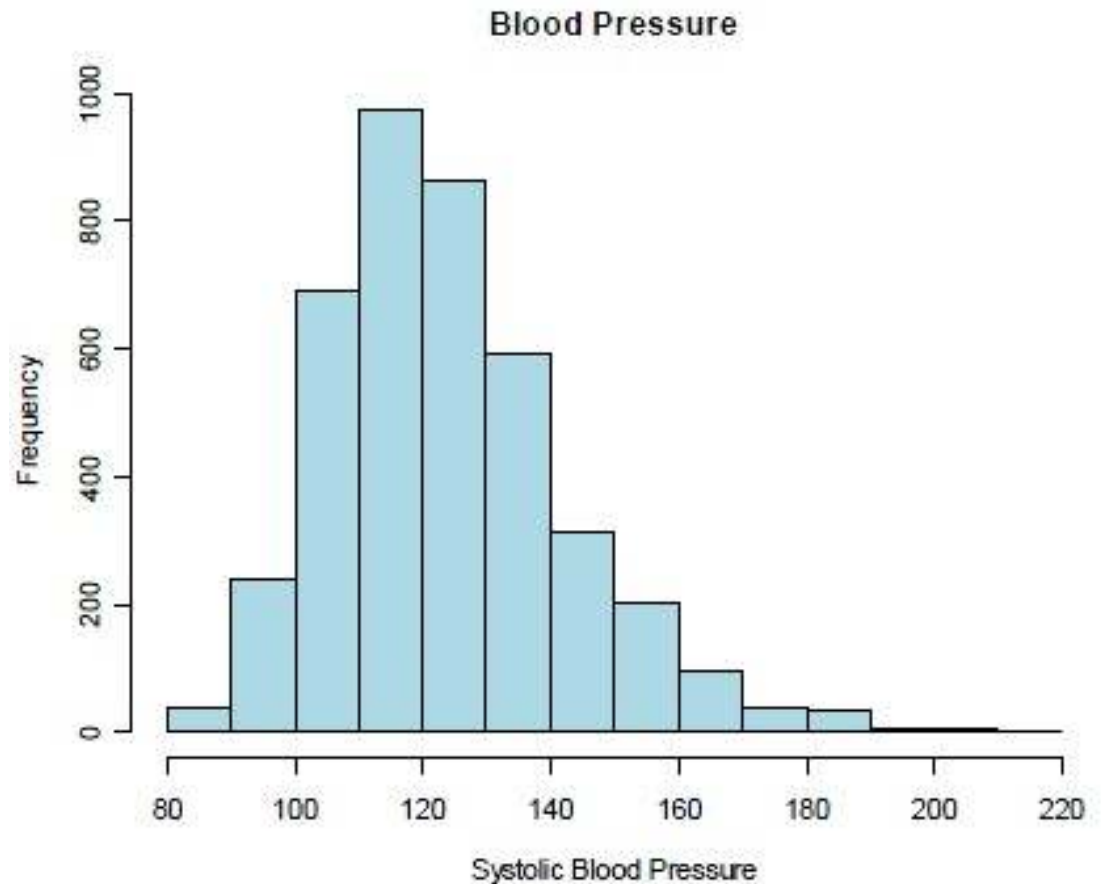
```
plot(dataset$Height)
```



```
plot(sort(dataset$Height))
```

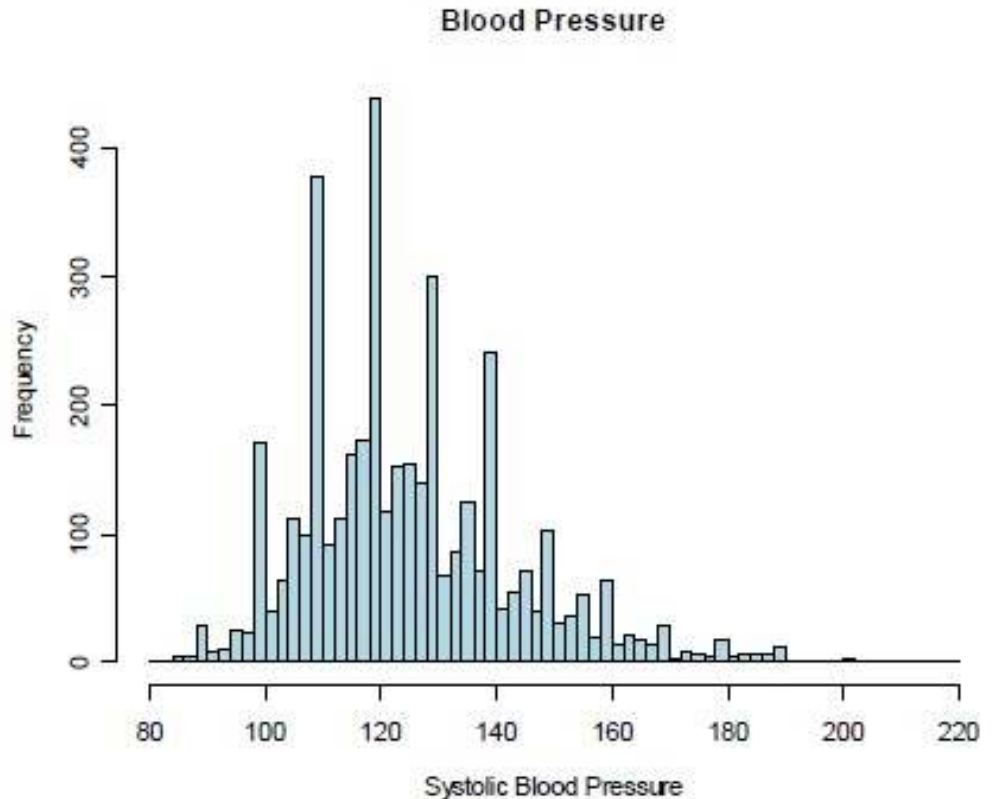
Histogram

```
hist(dataset$bp.sys, col = "lightblue",  
xlab = "Systolic Blood Pressure", main = "Blood Pressure")
```



Histogram

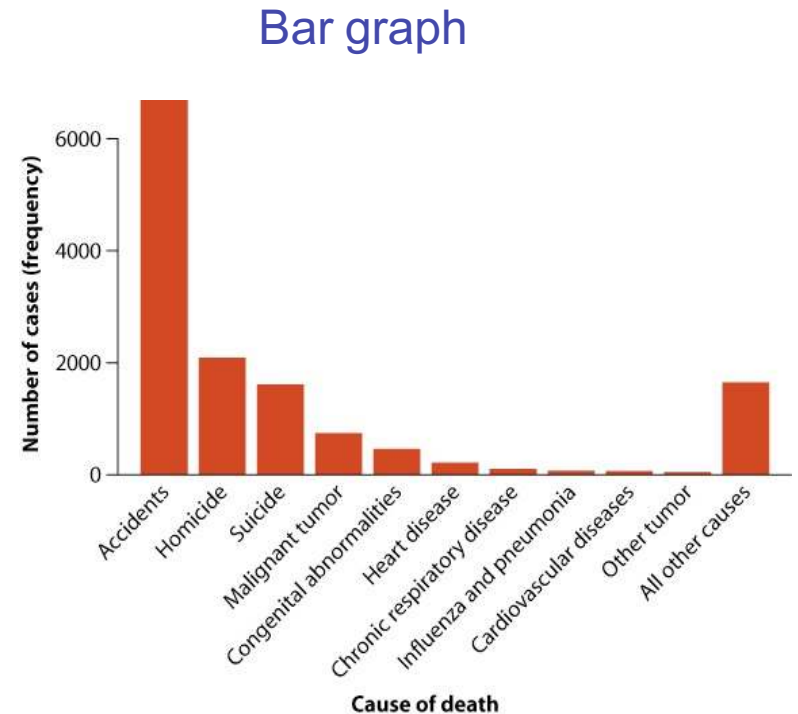
```
hist(dataset$bp.sys, col = "lightblue", breaks = seq(80,220,by=2),  
      xlab = "Systolic Blood Pressure", main = "Blood Pressure")
```



Bar graph

Cause of death	Frequency
Accidents	6,688
Homicide	2,093
Suicide	1,615
Malignant tumor	745
Heart disease	463
Congenital abnormalities	222
Chronic respiratory disease	107
Influenza and pneumonia	73
Cerebrovascular diseases	67
Other tumor	52
All other causes	1,653

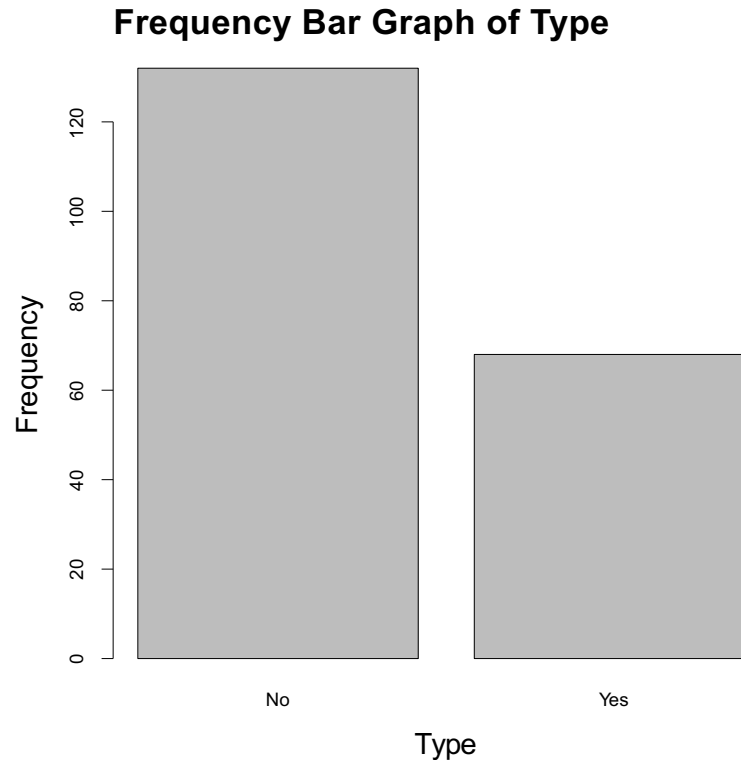
Frequency table showing the ten most common causes of death in Americans between 15 and 19 years of age in 1999. The total number of deaths is $n = 13,778$.



Bar graphs and frequencies

```
> type.freq <- table(Pima.tr$type)
```

```
> barplot(type.freq, xlab = "Type", ylab = "Frequency", main = "Frequency Bar Graph of Type")
```

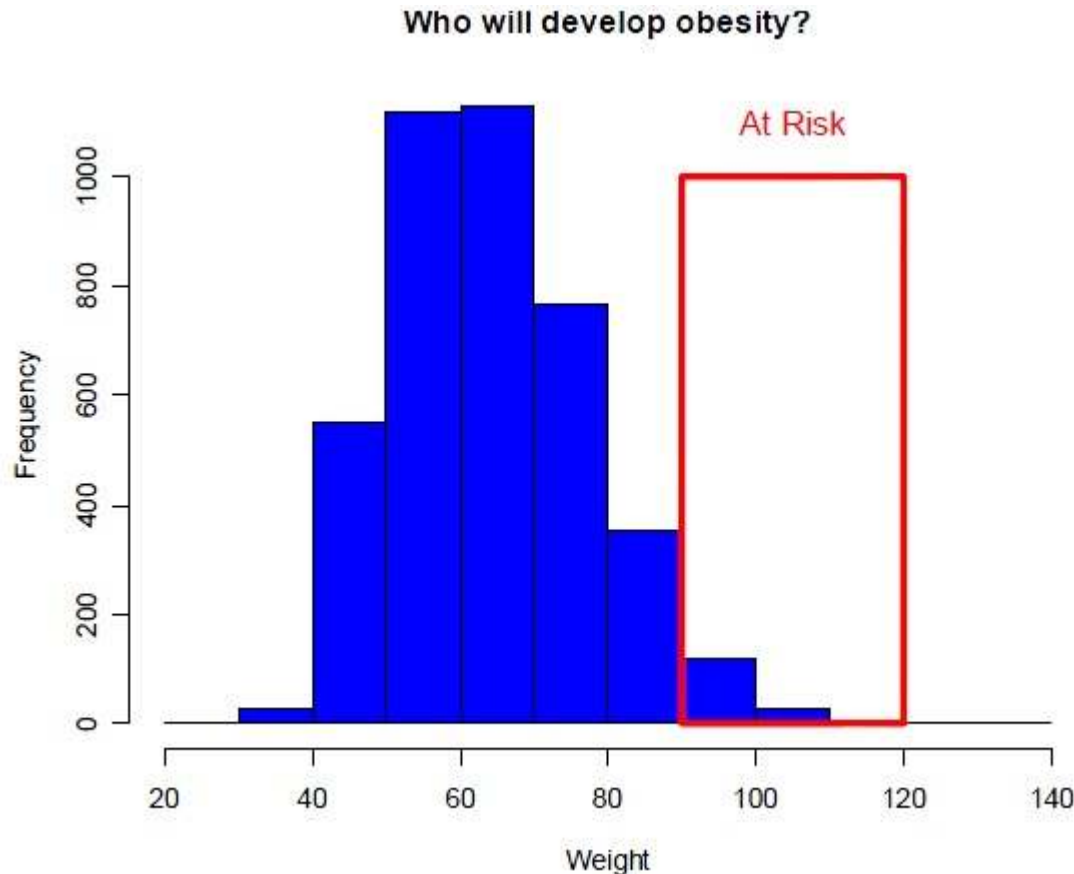


Adding a Label Inside a Plot

```
> hist(dataset$Weight, xlab = "Weight", main = "Who will develop obesity?",  
col = "blue")
```

```
> rect(90, 0, 120, 1000, border = "red", lwd = 4)
```

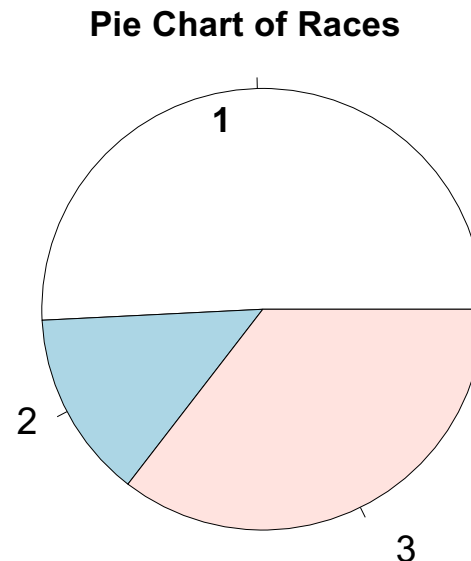
```
> text(105, 1100, "At Risk", col = "red", cex = 1.25)
```



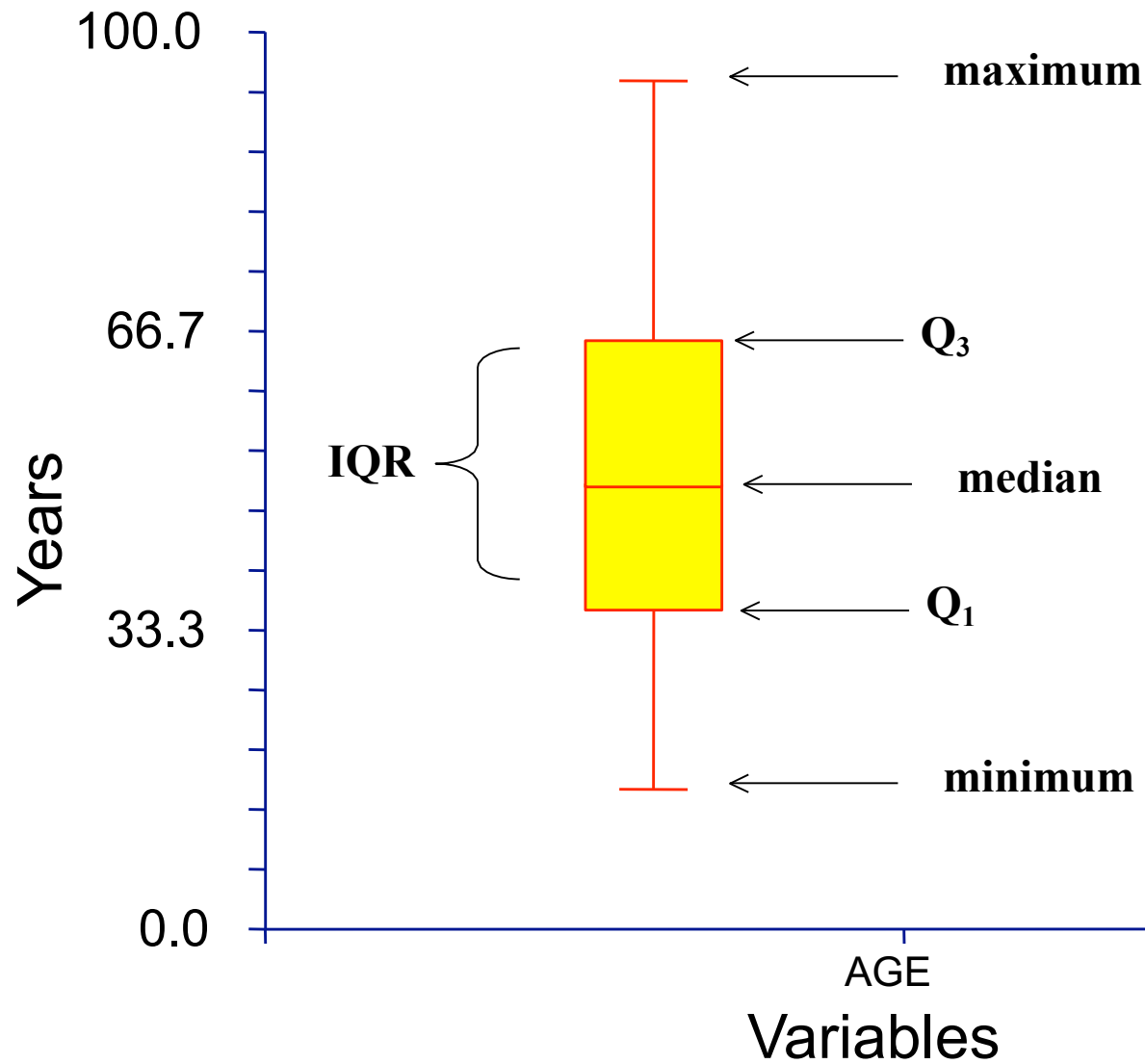
Pie chart

- We can use a pie chart to visualize the relative frequencies of different categories for a categorical variable.
- In a pie chart, the area of a circle is divided into sectors, each representing one of the possible categories of the variable.
- The area of each sector c is proportional to its frequency.

```
slices <- c(11, 4, 6)
lbls <- c("1", "2", "3",)
pie(slices, labels = lbls, main="Pie Chart of Races")
```

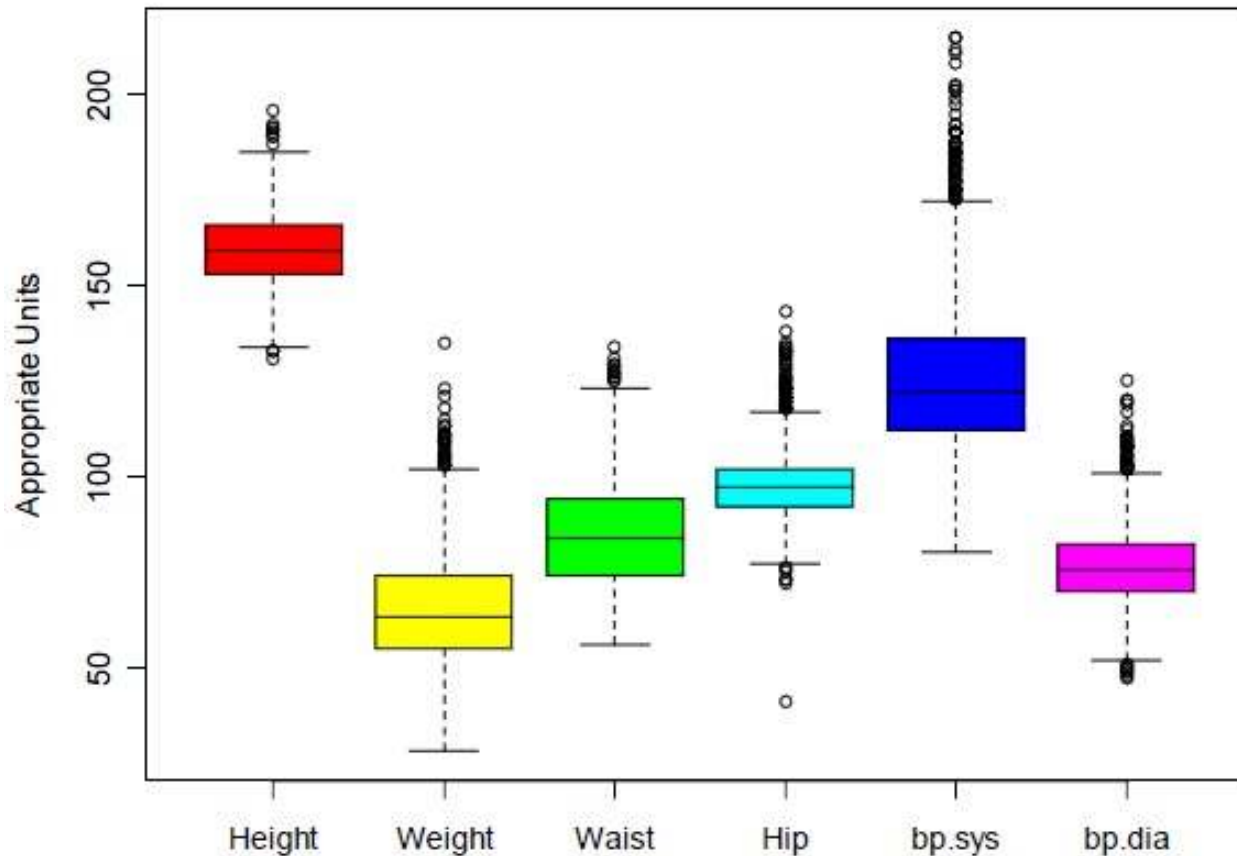


Box Plots



Example of Box Plots

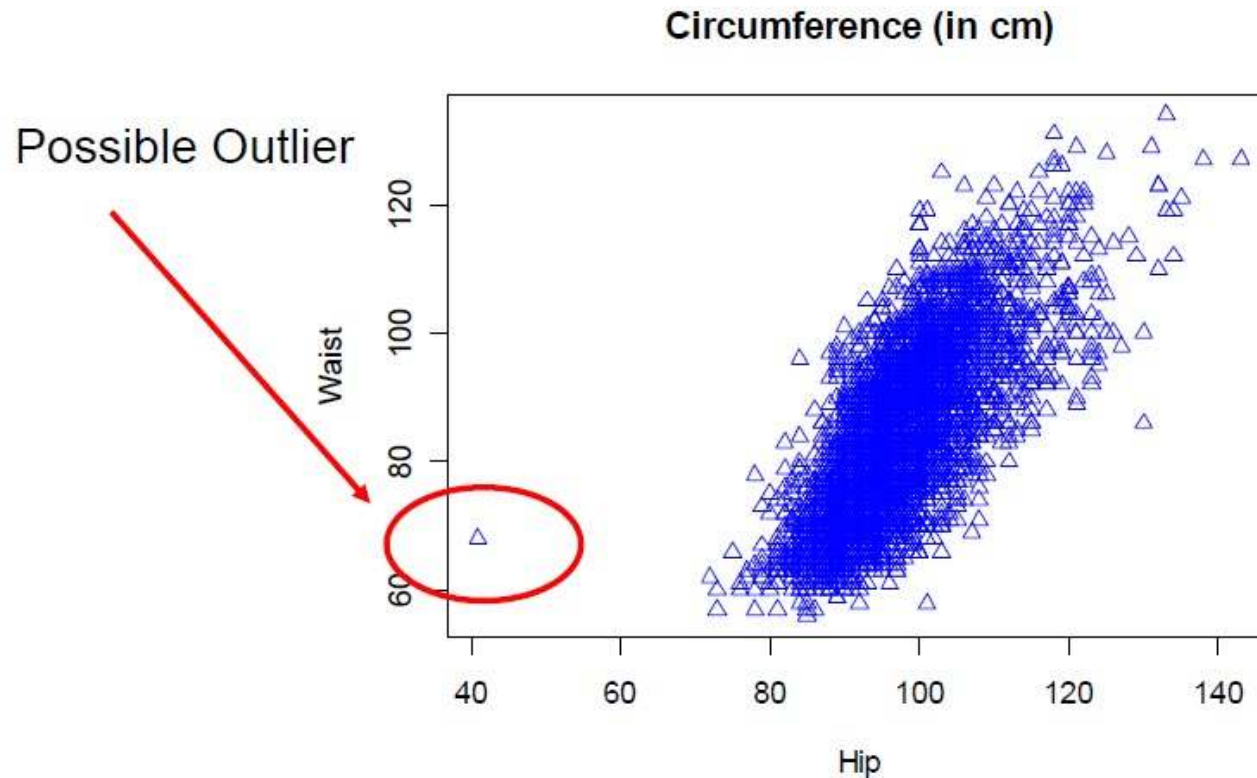
- Box plots can be used to compare attributes



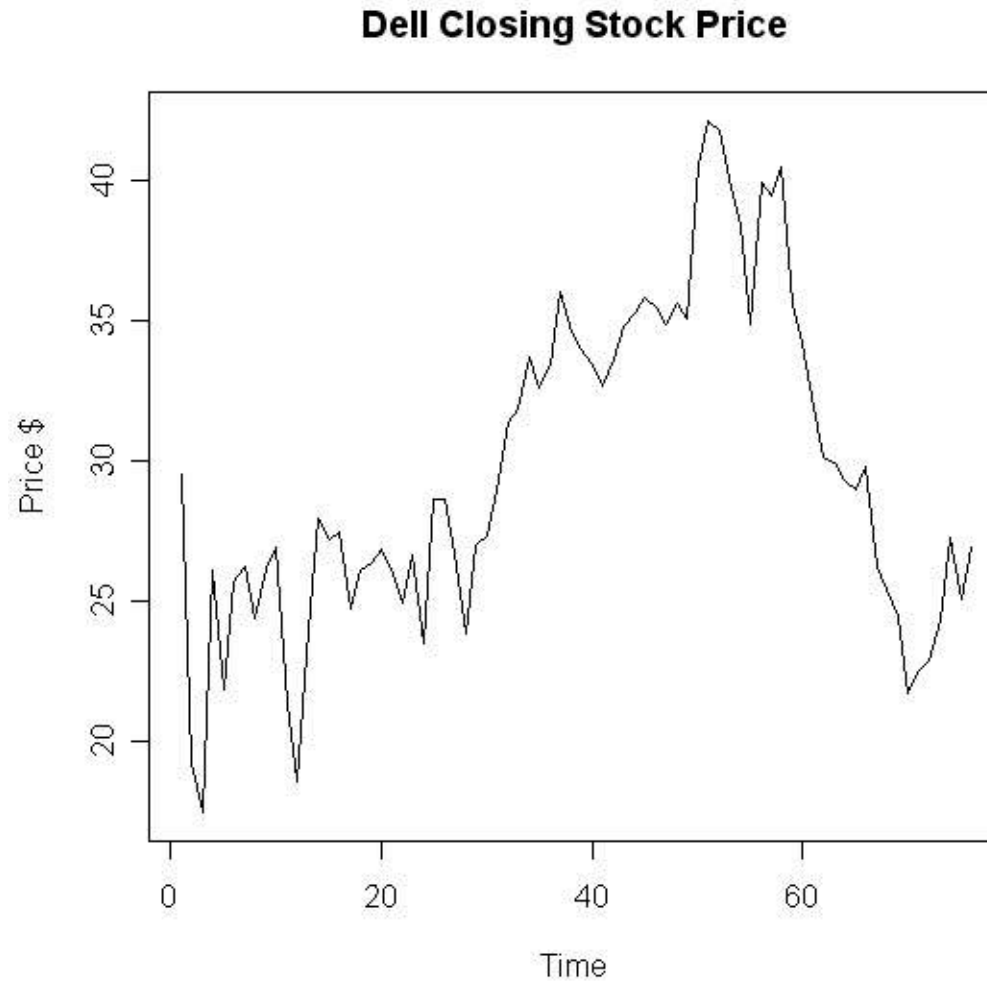
```
boxplot(dataset, col = rainbow(6), ylab = "Appropriate Units")
```

Scatter plots -Plotting Two Vectors

```
plot(dataset$Hip, dataset$Waist,  
      xlab = "Hip", ylab = "Waist",  
      main = "Circumference (in cm)", pch = 2, col = "blue")
```



Line Plots



```
plot(t1,D2$DELL,type="l",main='Dell Closing Stock Price',  
xlab='Time',ylab='Price $'))
```

Adding a Legend



```
legend(60,45,c('Intel','Dell'),lty=c(1,2))
```

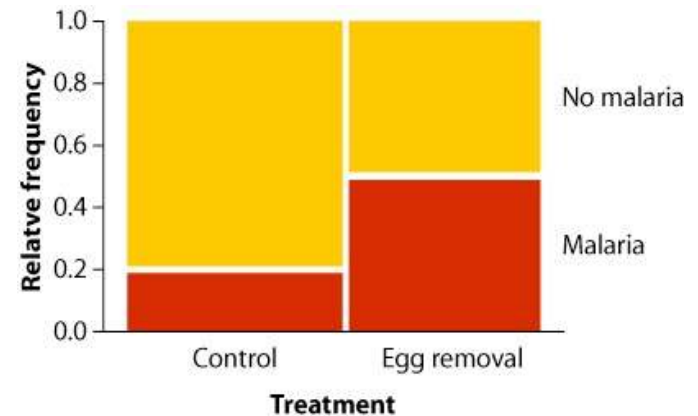
Mosaic plot

Association between reproductive effort and avian malaria

Table 2.3A. Contingency table showing incidence of malaria in female great tits subjected to experimental egg removal.



	control group	egg removal group	row total
malaria	7	15	22
no malaria	28	15	43
column total	35	30	65

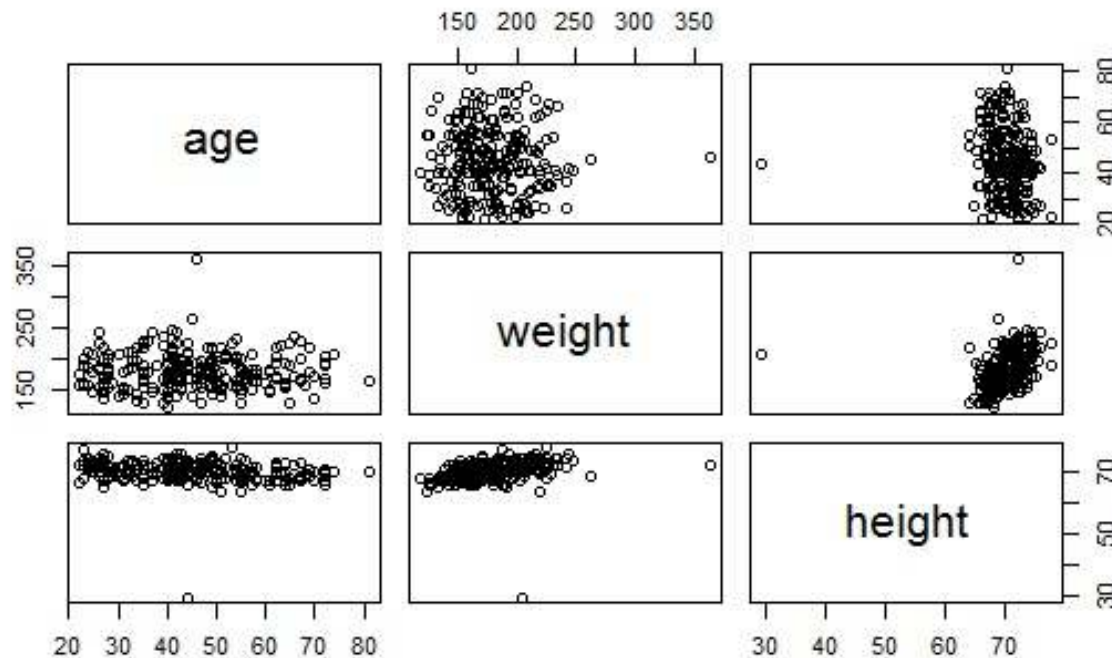


```
>library(vcd)
```

```
>mosaic(HairEyeColor, shade=TRUE, legend=FALSE)
```

Plotting Contents of a Dataset as Matrix

```
>plot(dataset[c(5,6,7)])
```

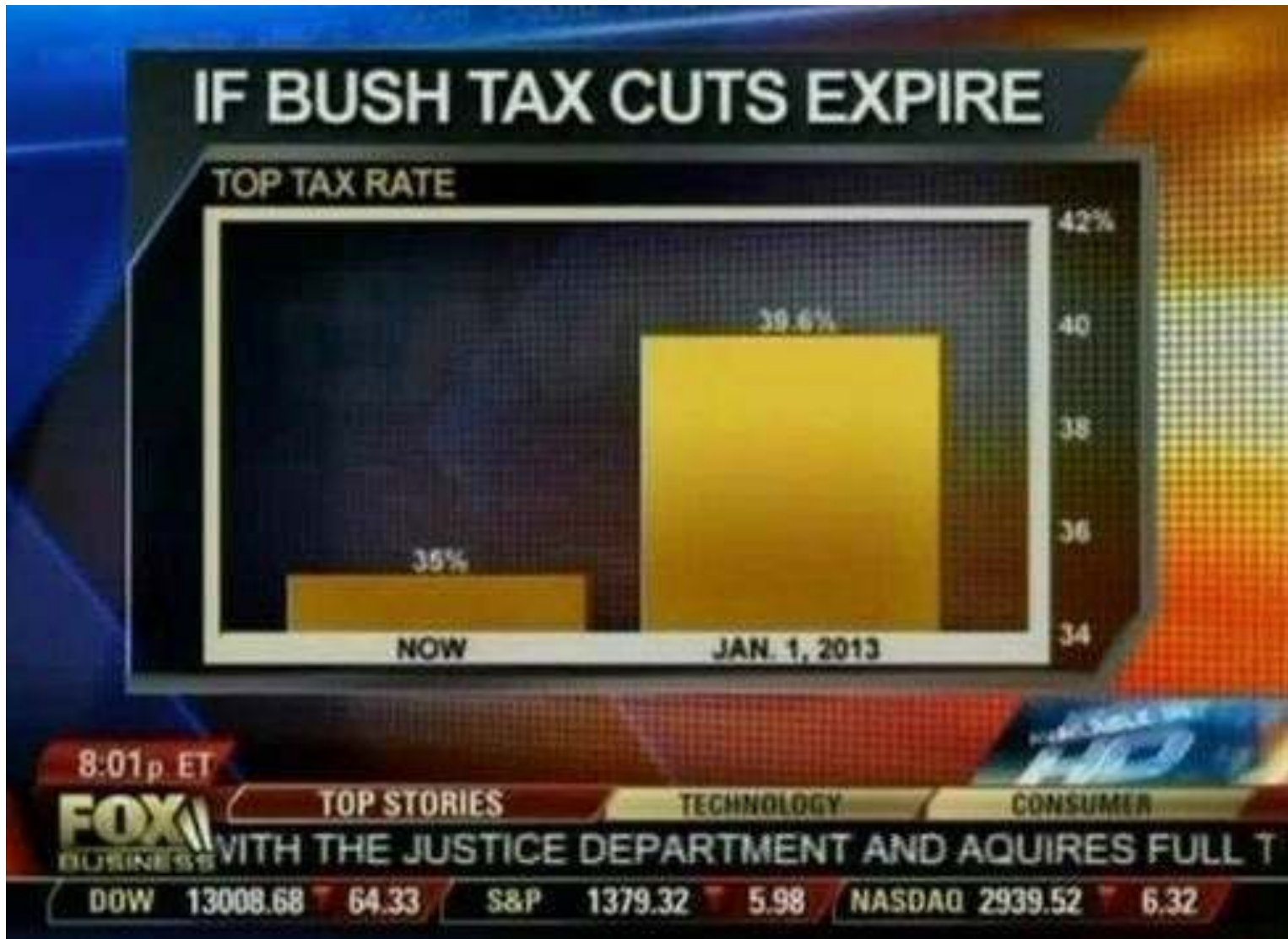


Effective Visualizations

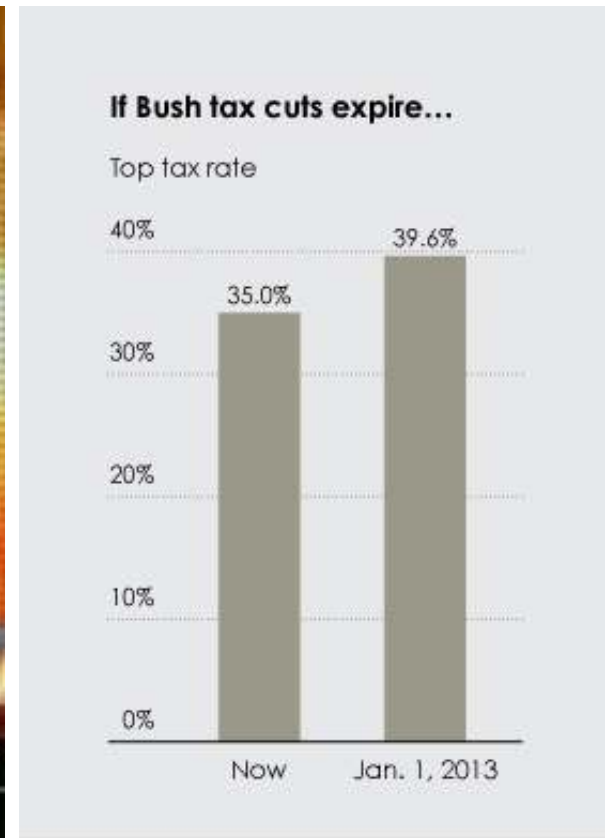
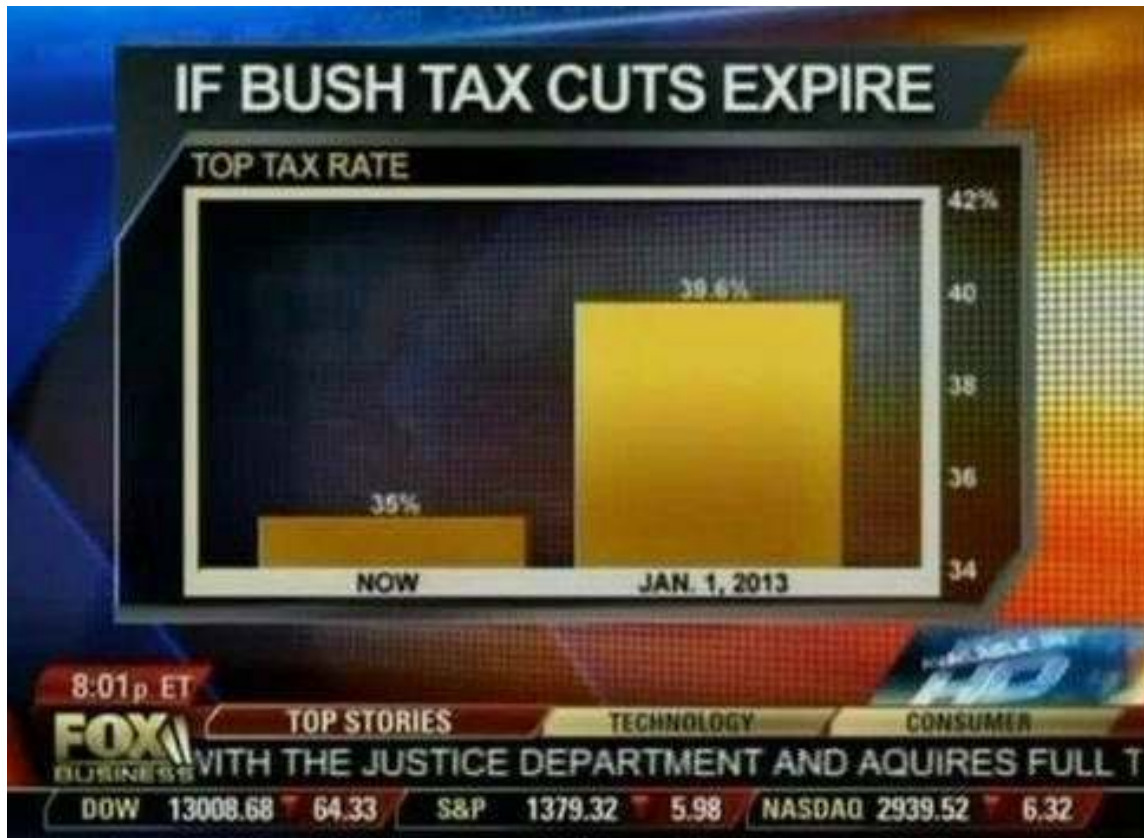
1. Have graphical integrity
2. Keep it simple
3. Use the right display
4. Use color strategically
5. Tell a story with data

Graphical Integrity

Graphical Integrity



Scale Distortions



JOB LOSS BY QUARTER

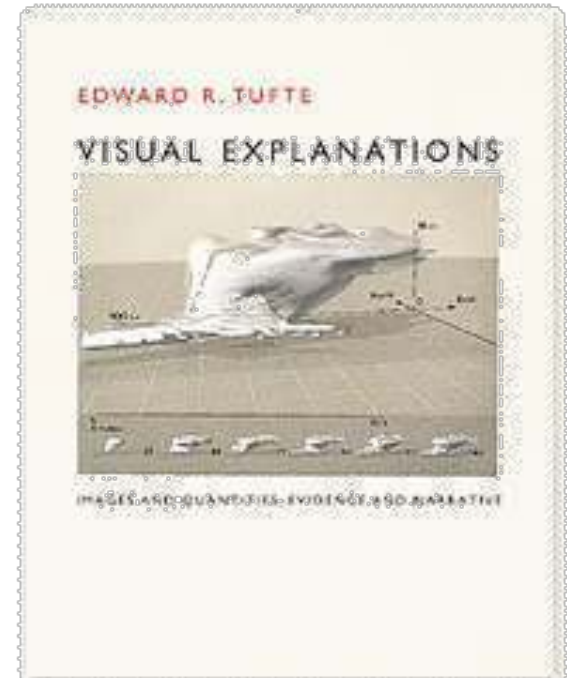
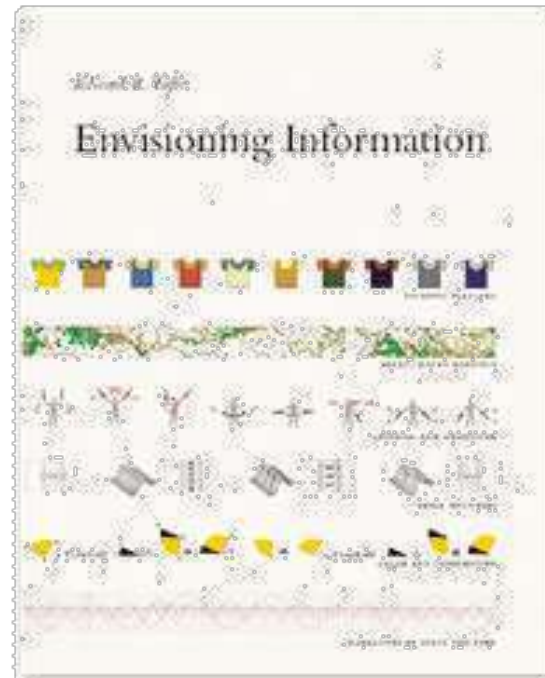
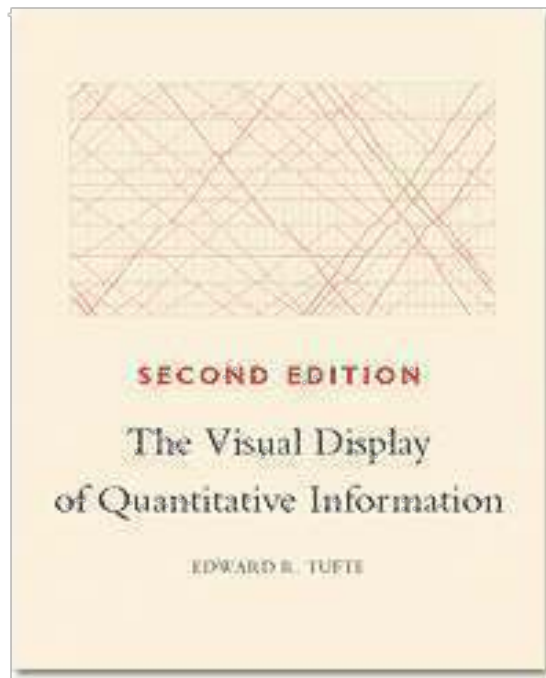


Scale Distortions



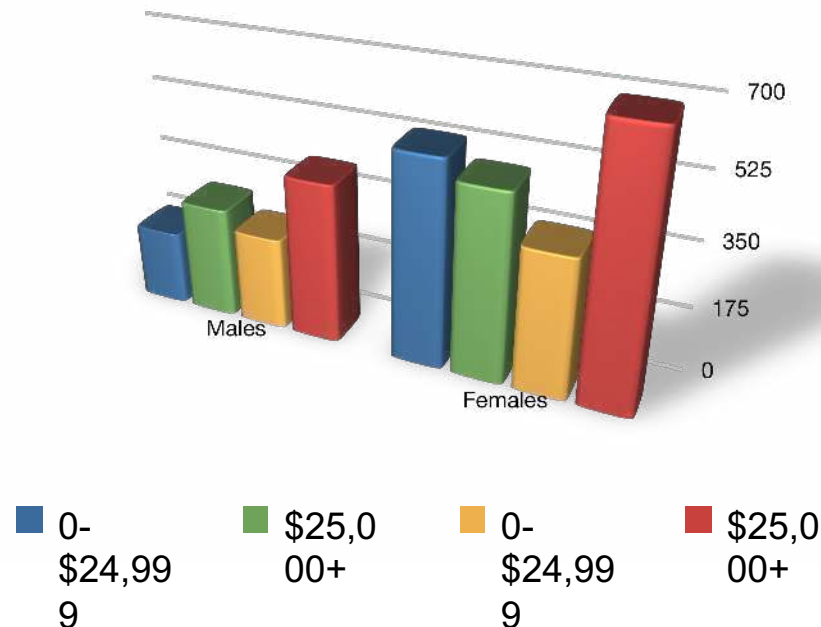
Keep It Simple

Edward Tufte



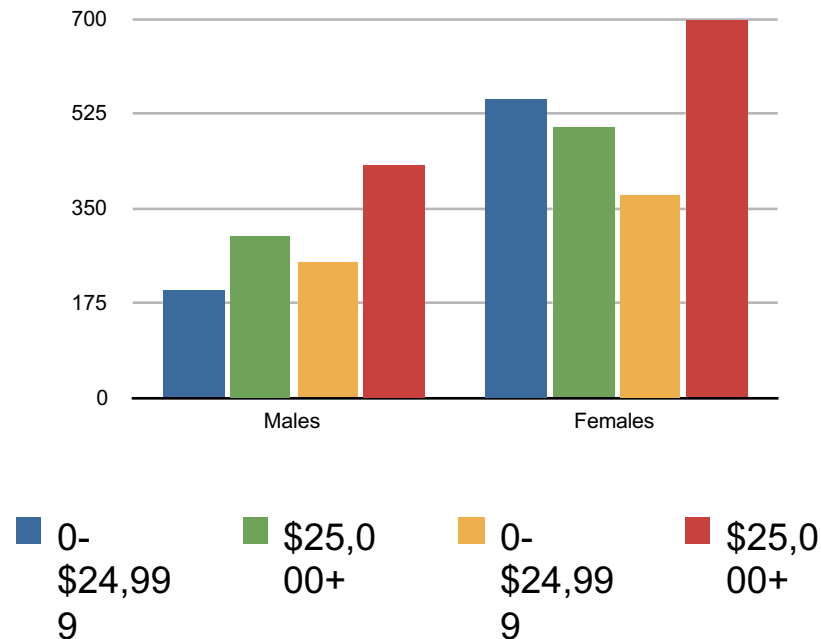
Maximize Data-Ink Ratio

$$\text{Data-Ink Ratio} = \frac{\text{Data ink}}{\text{Total ink used in graphic}}$$

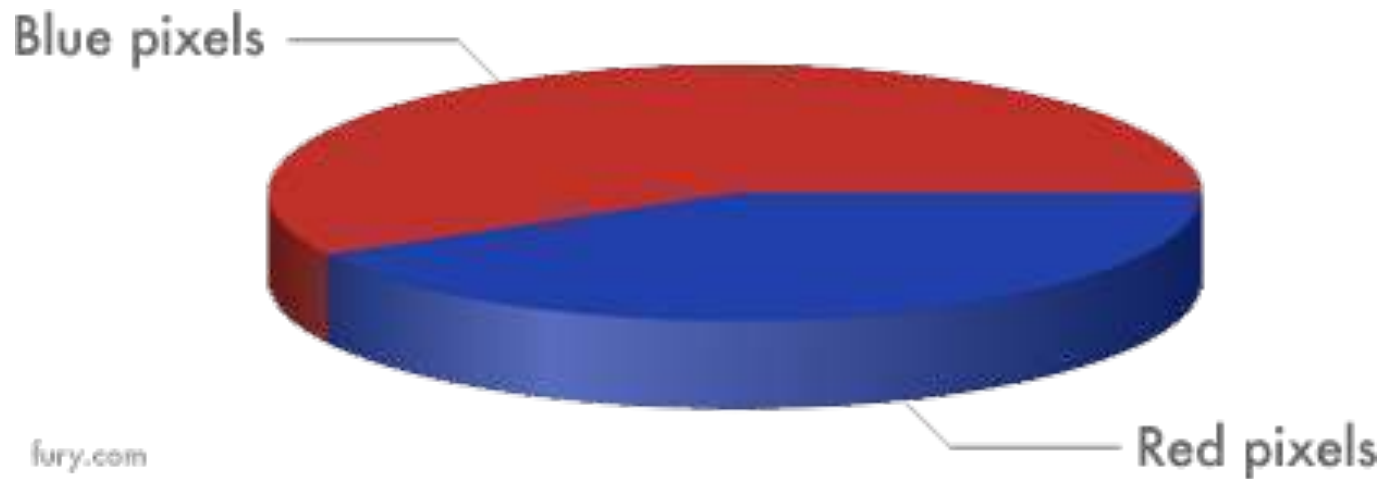


Maximize Data-Ink Ratio

$$\text{Data-Ink Ratio} = \frac{\text{Data ink}}{\text{Total ink used in graphic}}$$

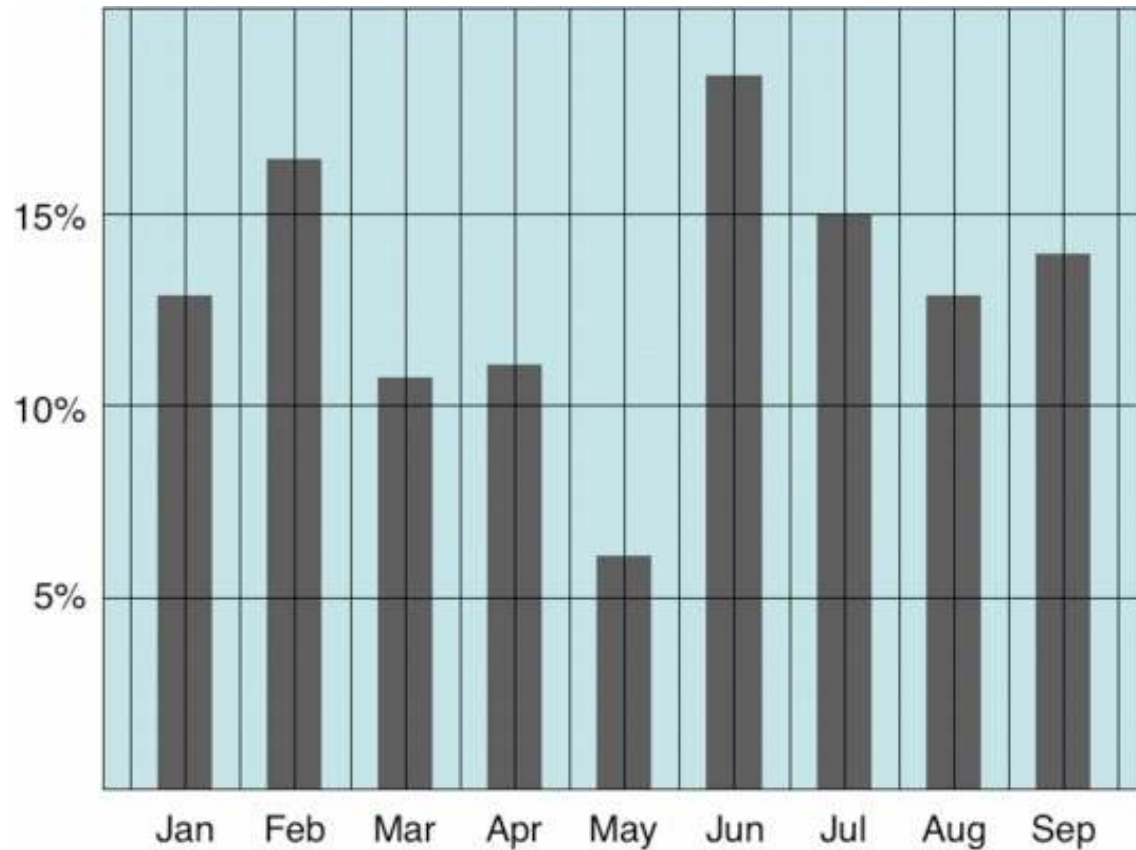


Why 3D pie charts are bad

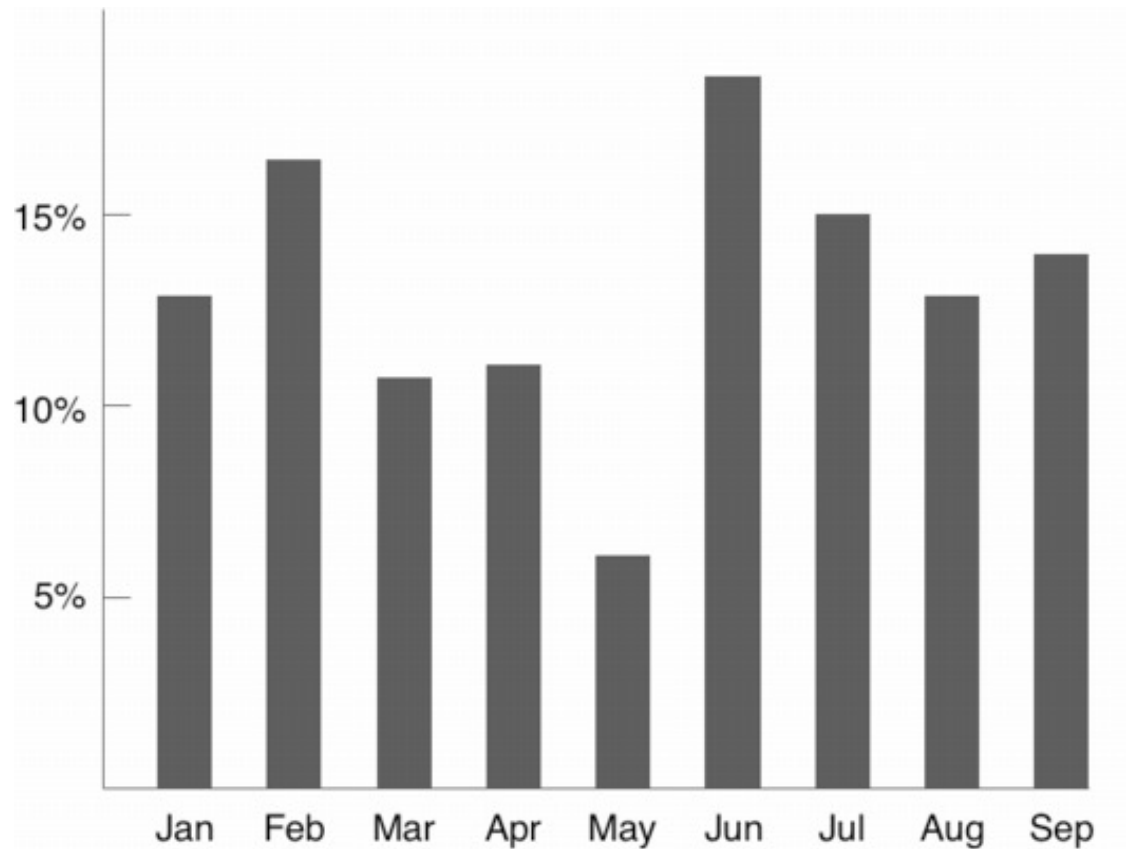


Avoid Chartjunk

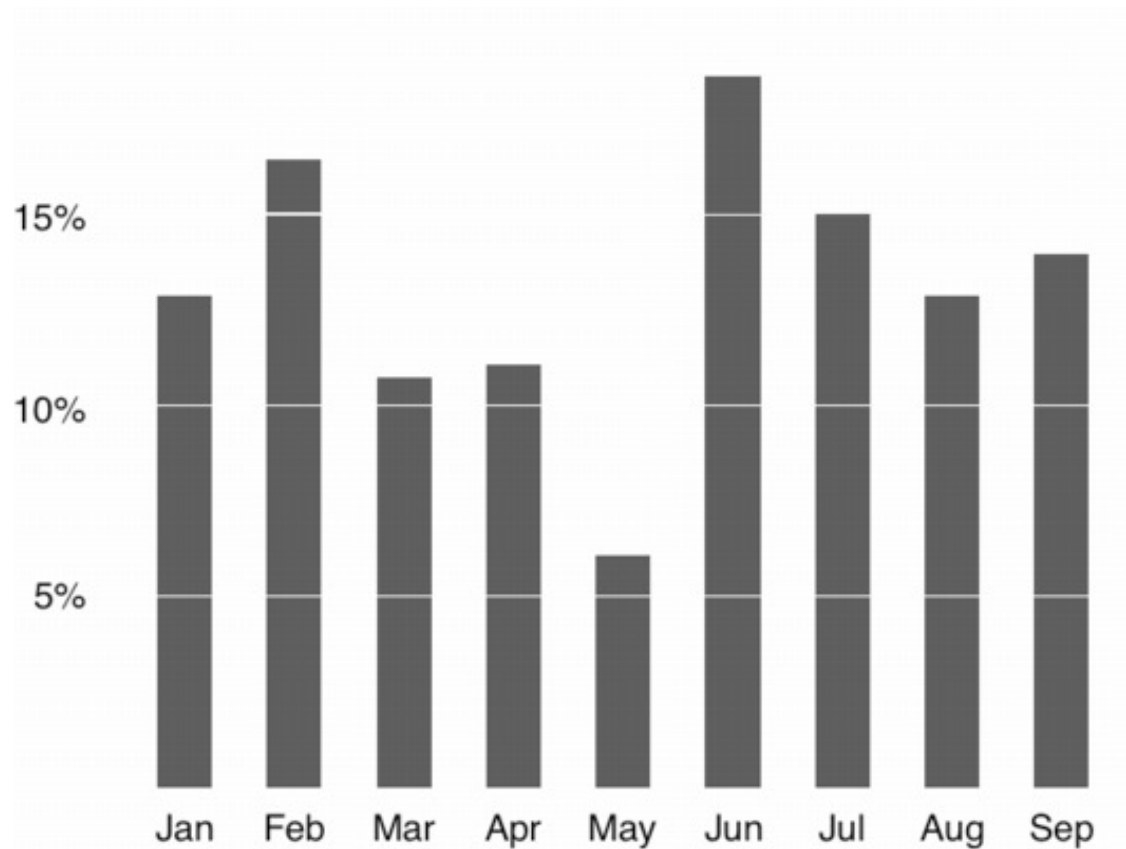
Extraneous visual elements that distract from the message



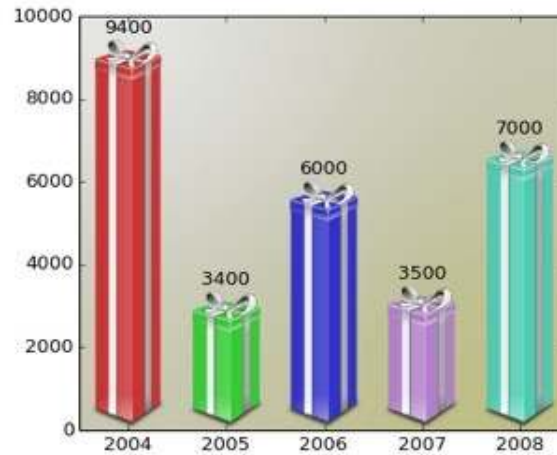
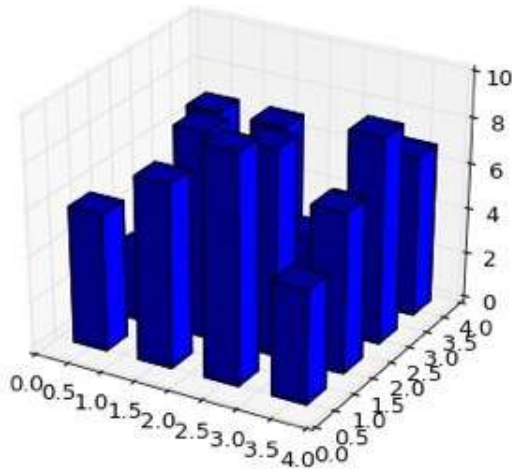
Avoid Chartjunk



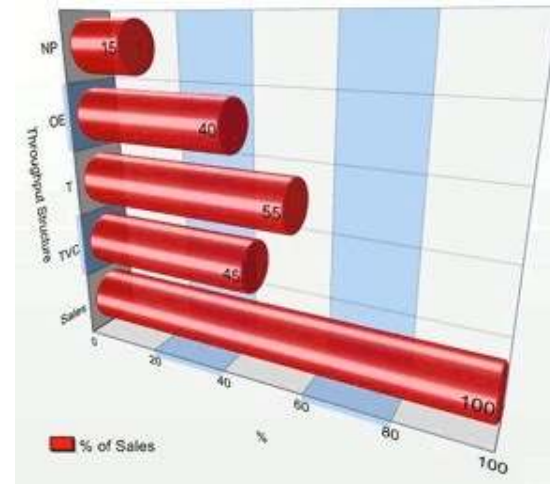
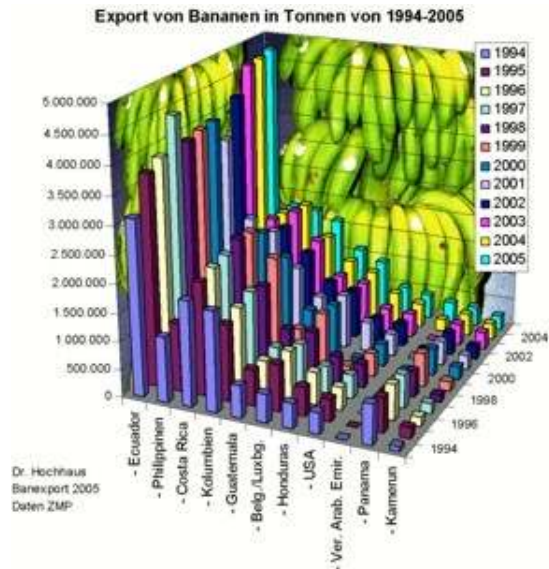
Avoid Chartjunk



Don't!



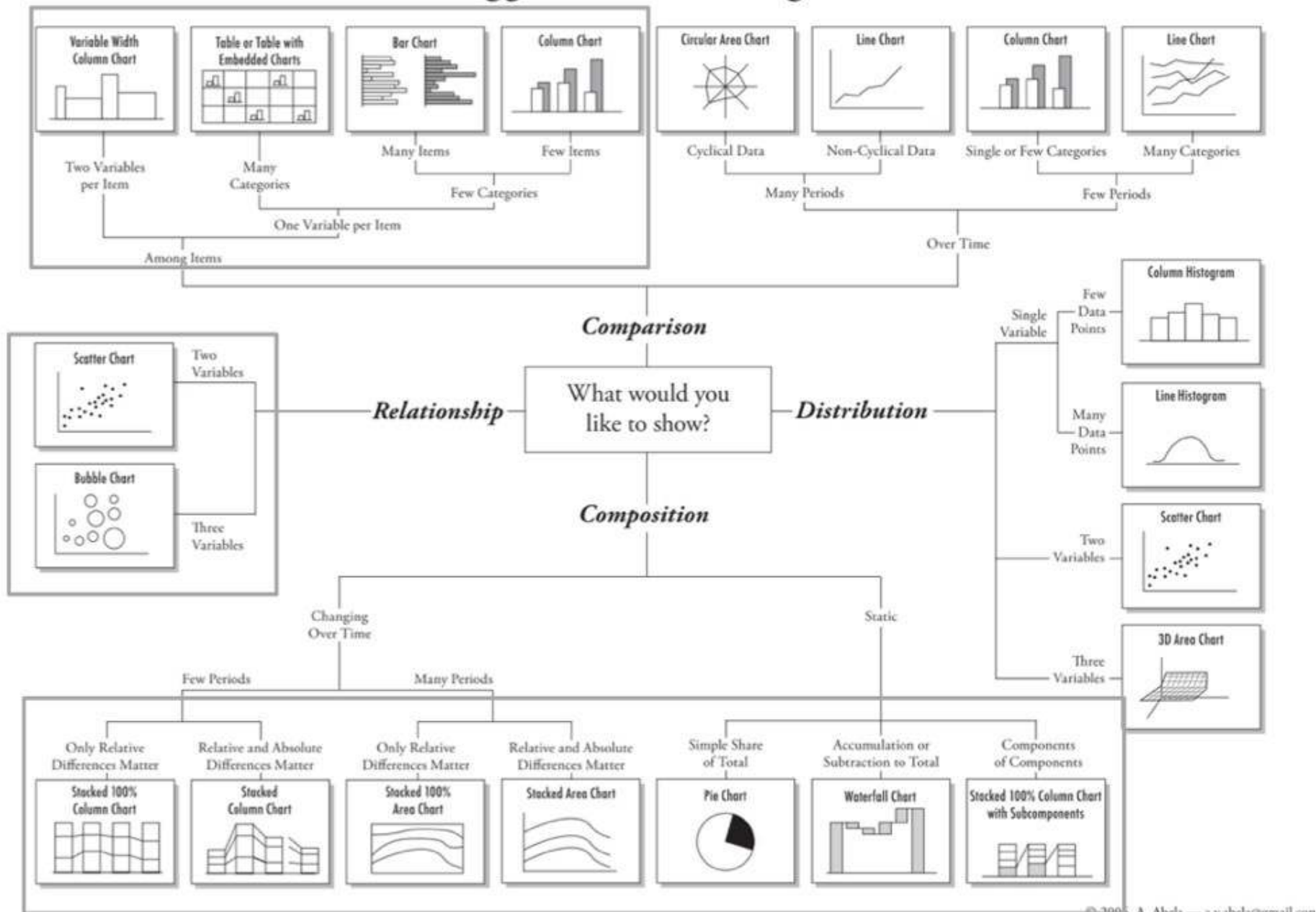
matplotlib
gallery



Excel Charts
Blog

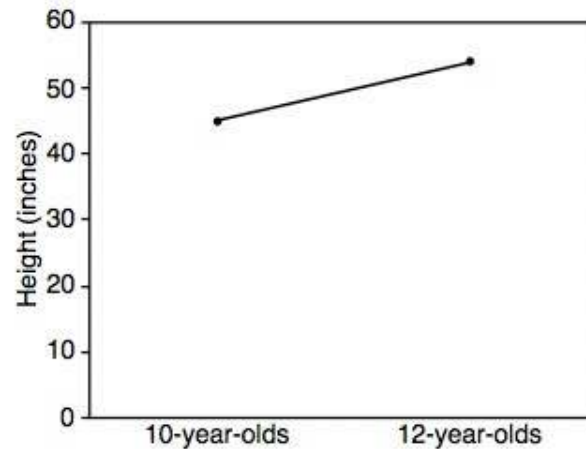
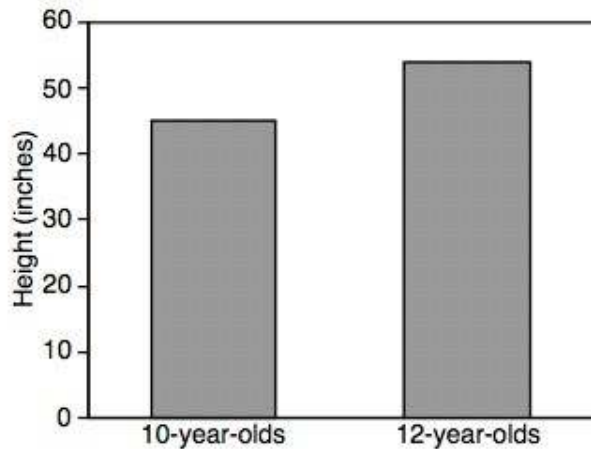
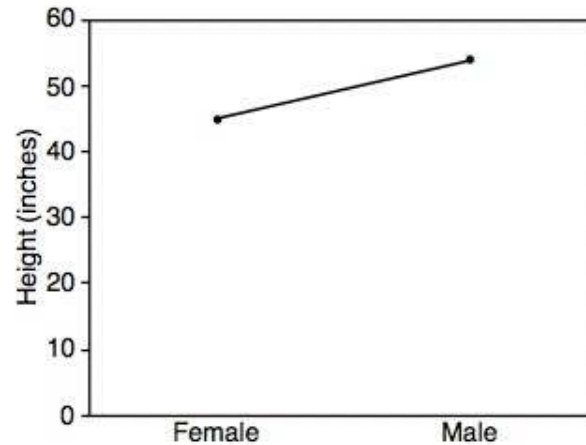
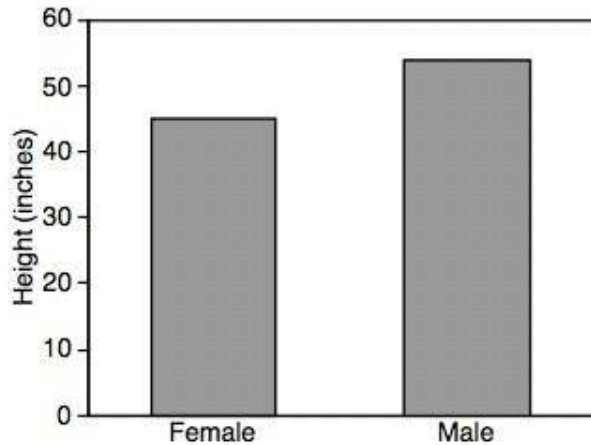
Use The Right Display

Chart Suggestions—A Thought-Starter



Comparisons

Bars vs. Lines



Trends

Apple Inc. (AAPL) - NasdaqGS

[+ Add to Portfolio](#)

[Like](#)

6k

601.10 ↑ 15.53 (2.65%) 4:00PM EDT | After Hours: **604.60** ↑ 3.50 (0.58%) 7:15PM EDT - Nasdaq Real Time Price

Enter name(s) or symbol(s)

GET CHART

COMPARE

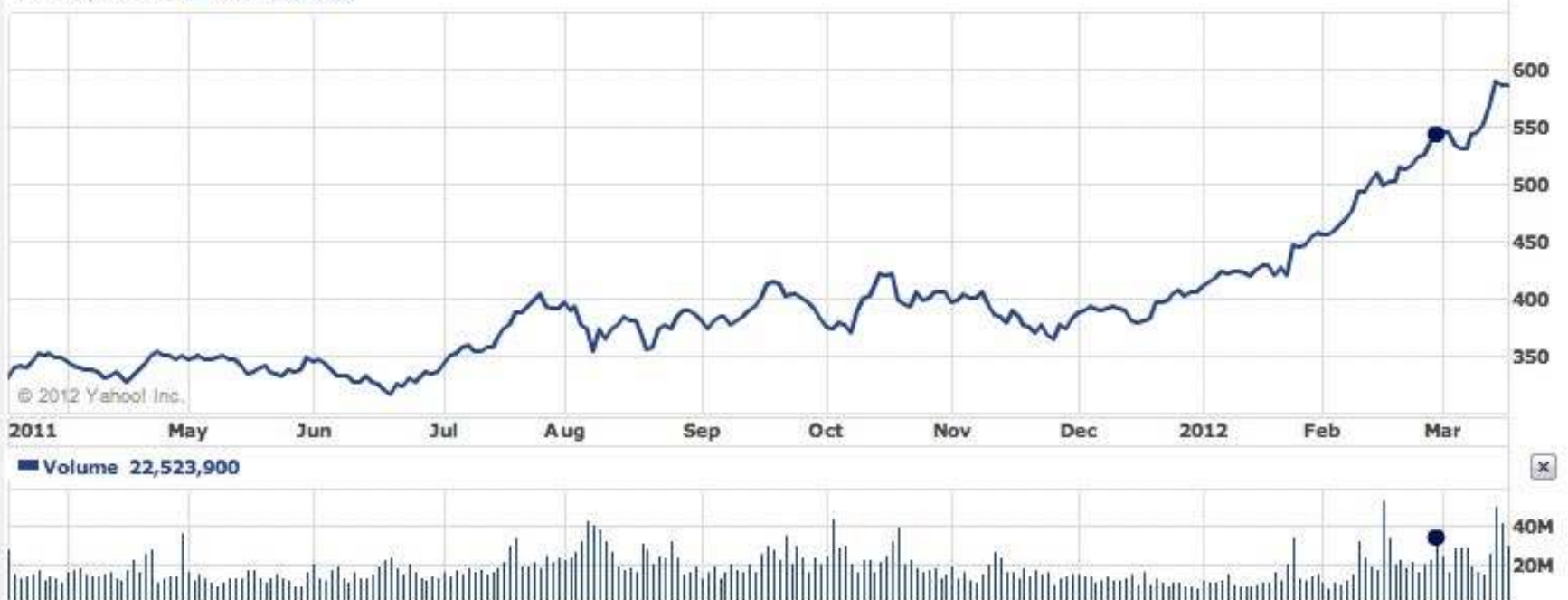
EVENTS ▾

TECHNICAL INDICATORS ▾

CHART SETTINGS ▾

RESET

Feb 10, 2012 : AAPL 493.42



1D 5D 1M YTD 3M 6M 1Y 2Y 5Y Max

FROM: Mar 18 2011 TO: Mar 16 2012

1984

1989

1994

1999

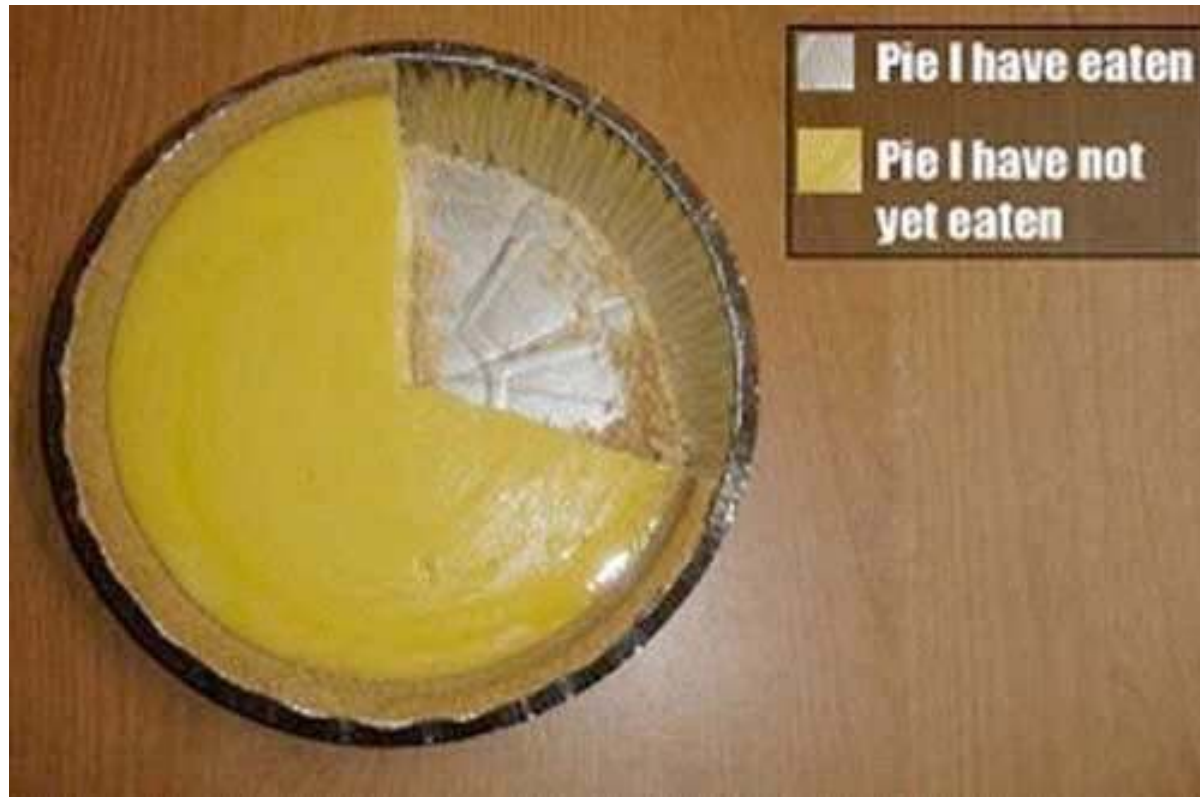
2004

2009

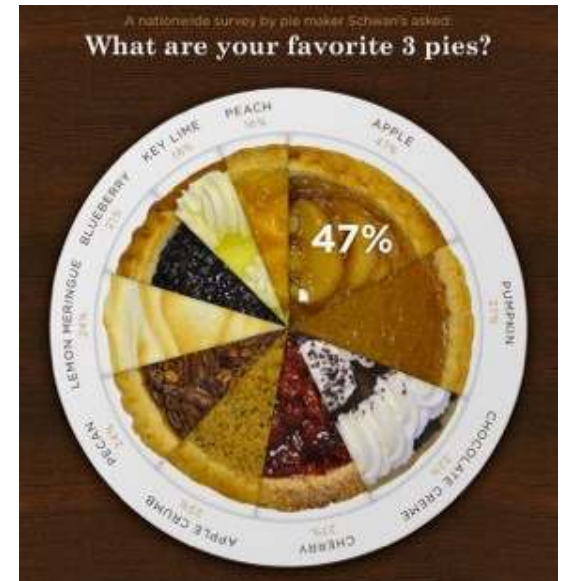
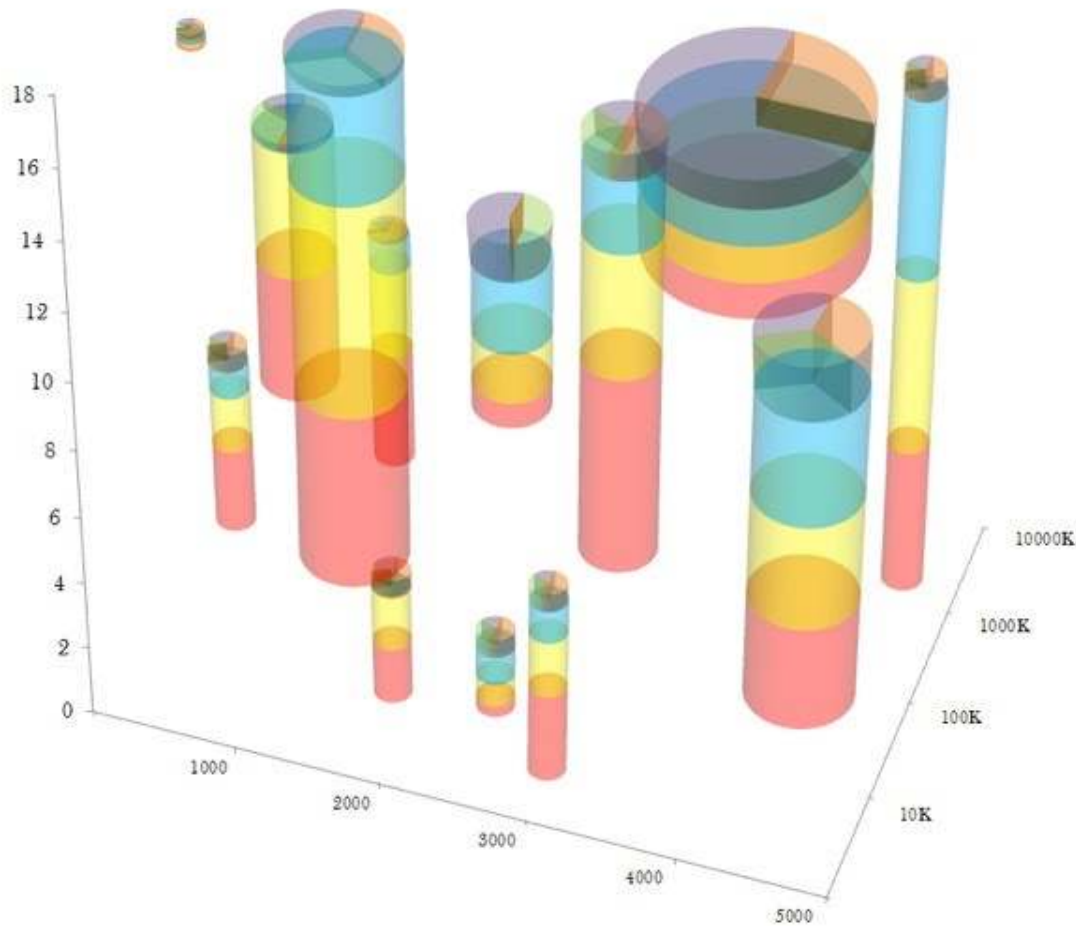
[Basic Chart](#) | [Full Screen](#) | [Print](#) | [Share](#) | [Send Feedback](#)

Proportions

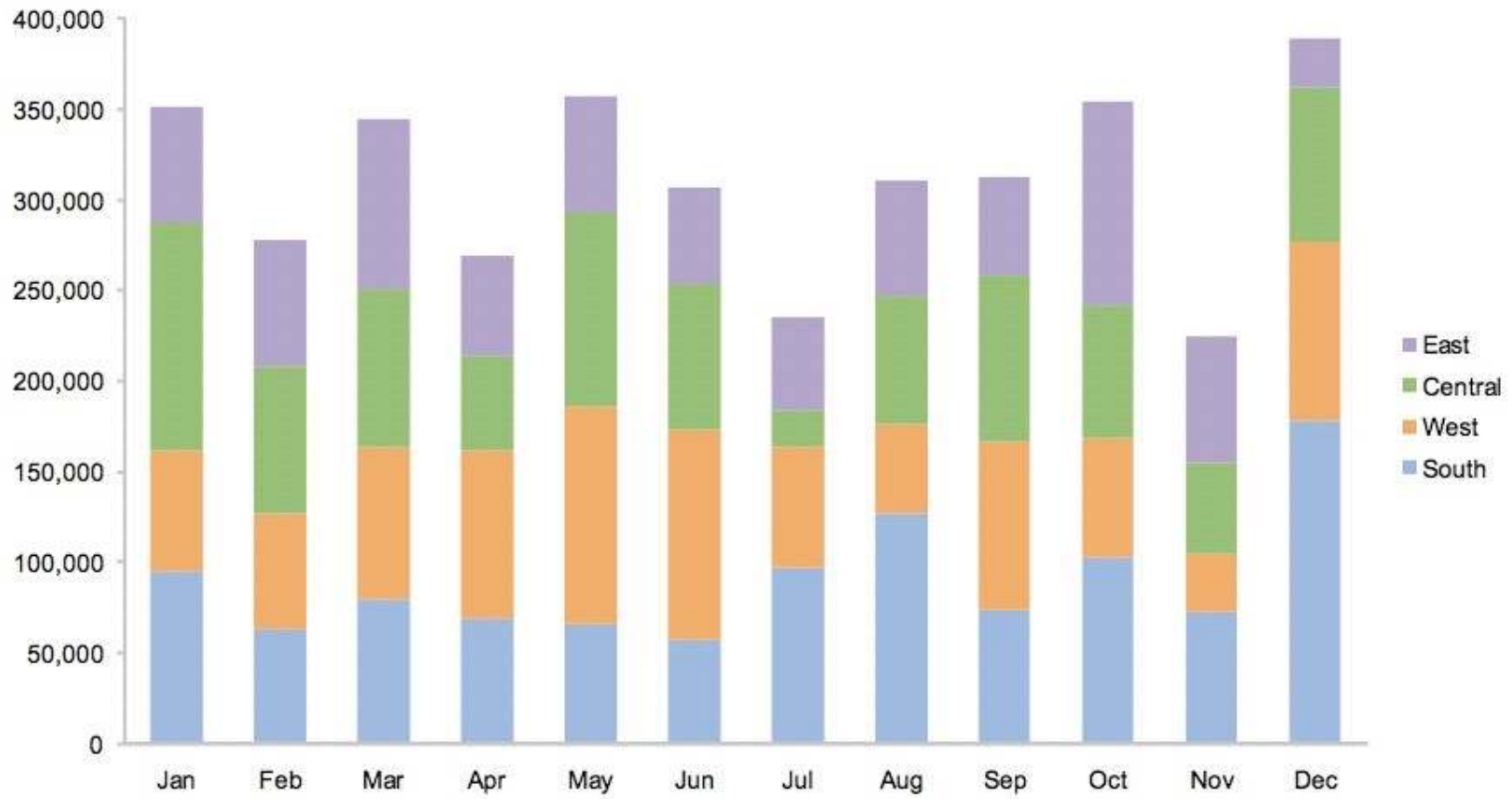
Pie Charts



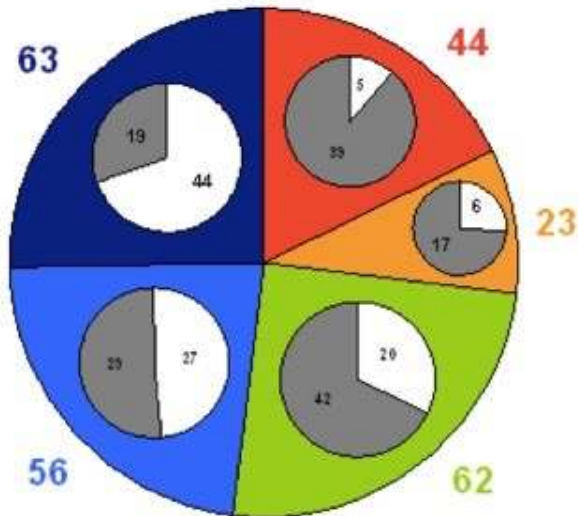
eagerpies.com



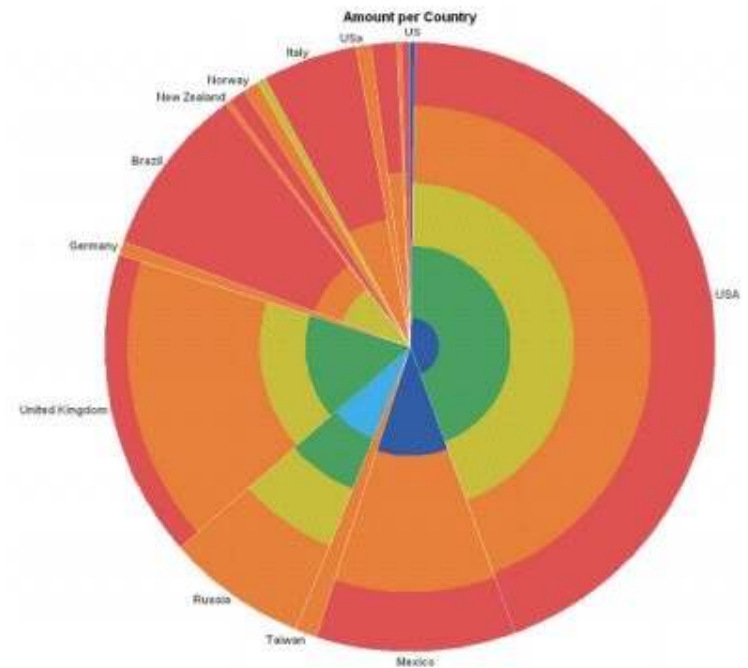
Stacked Bar Chart



Don't!

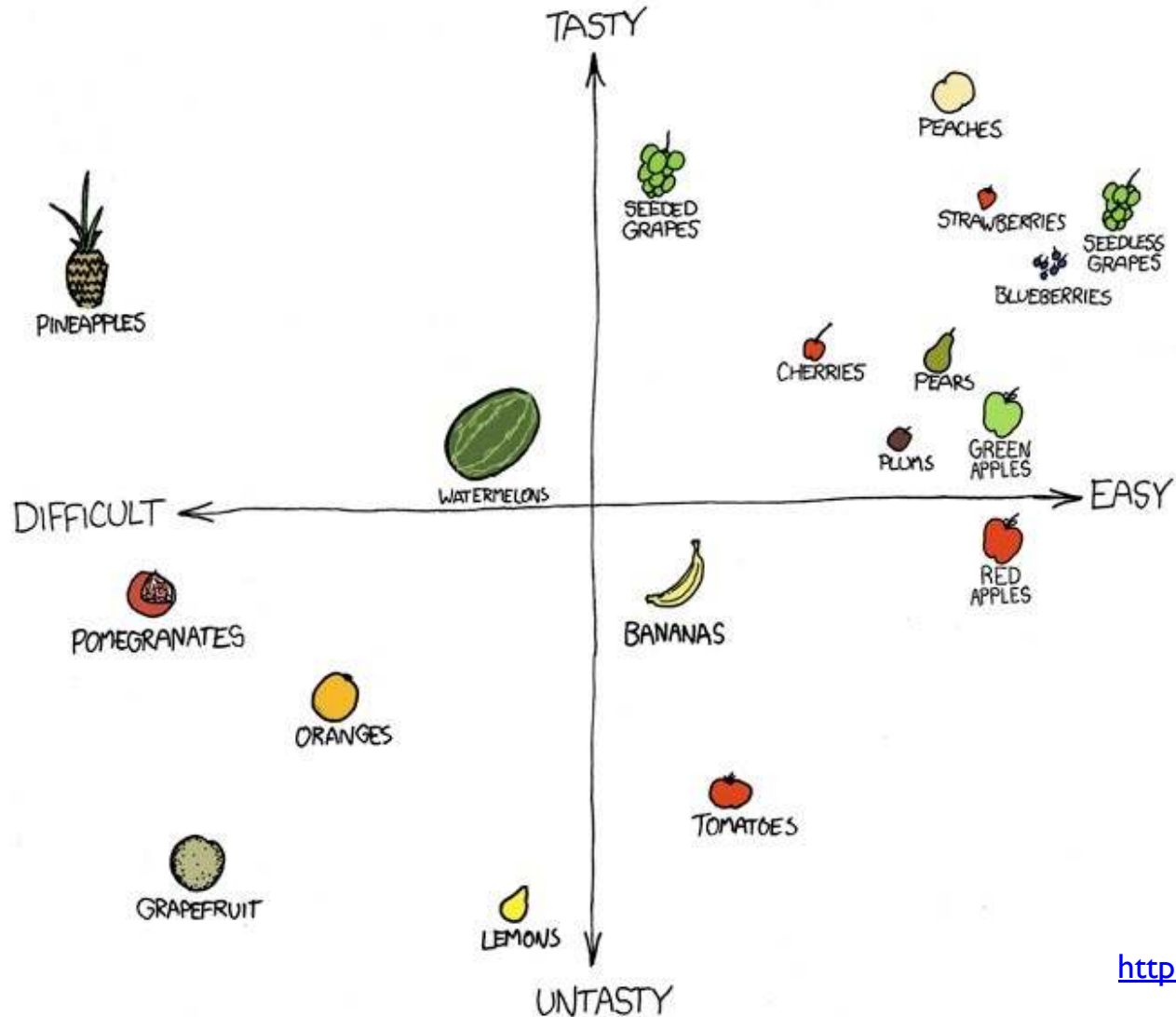


- new folds
- new folds, partial similarity
- putative analogs
- putative homologs
- recognizable homologs
- hypothesis about function
- no hypothesis about function

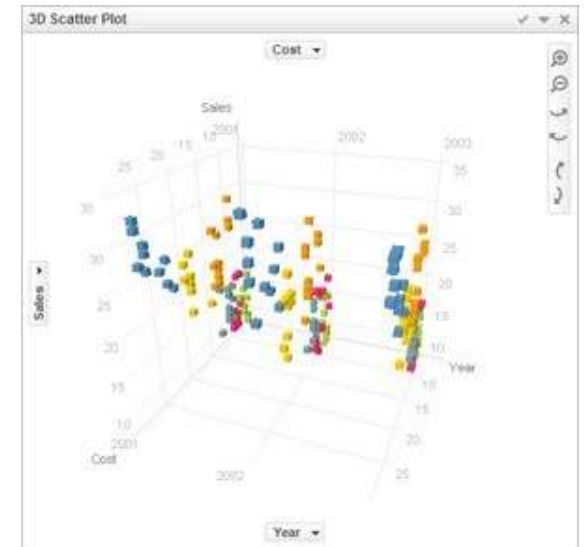
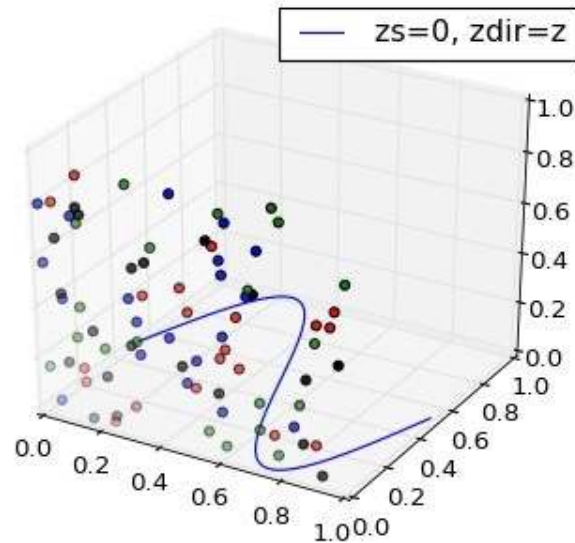
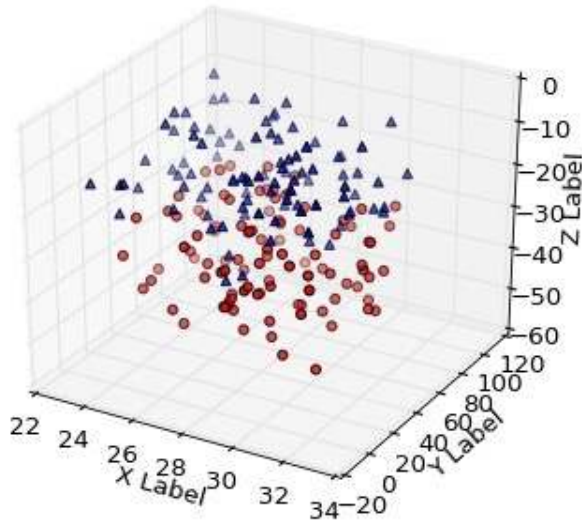


Correlations

Scatterplots

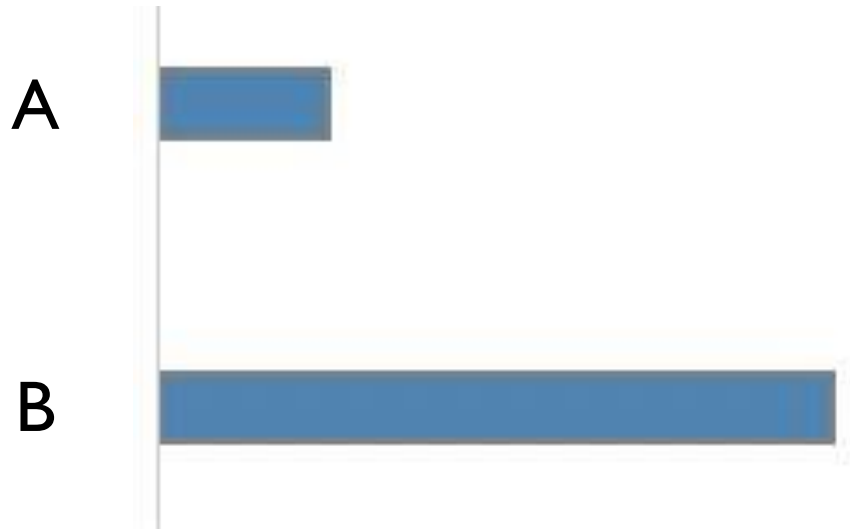


Don't!



Perceptual Effectiveness

How much longer?



How much steeper slope?

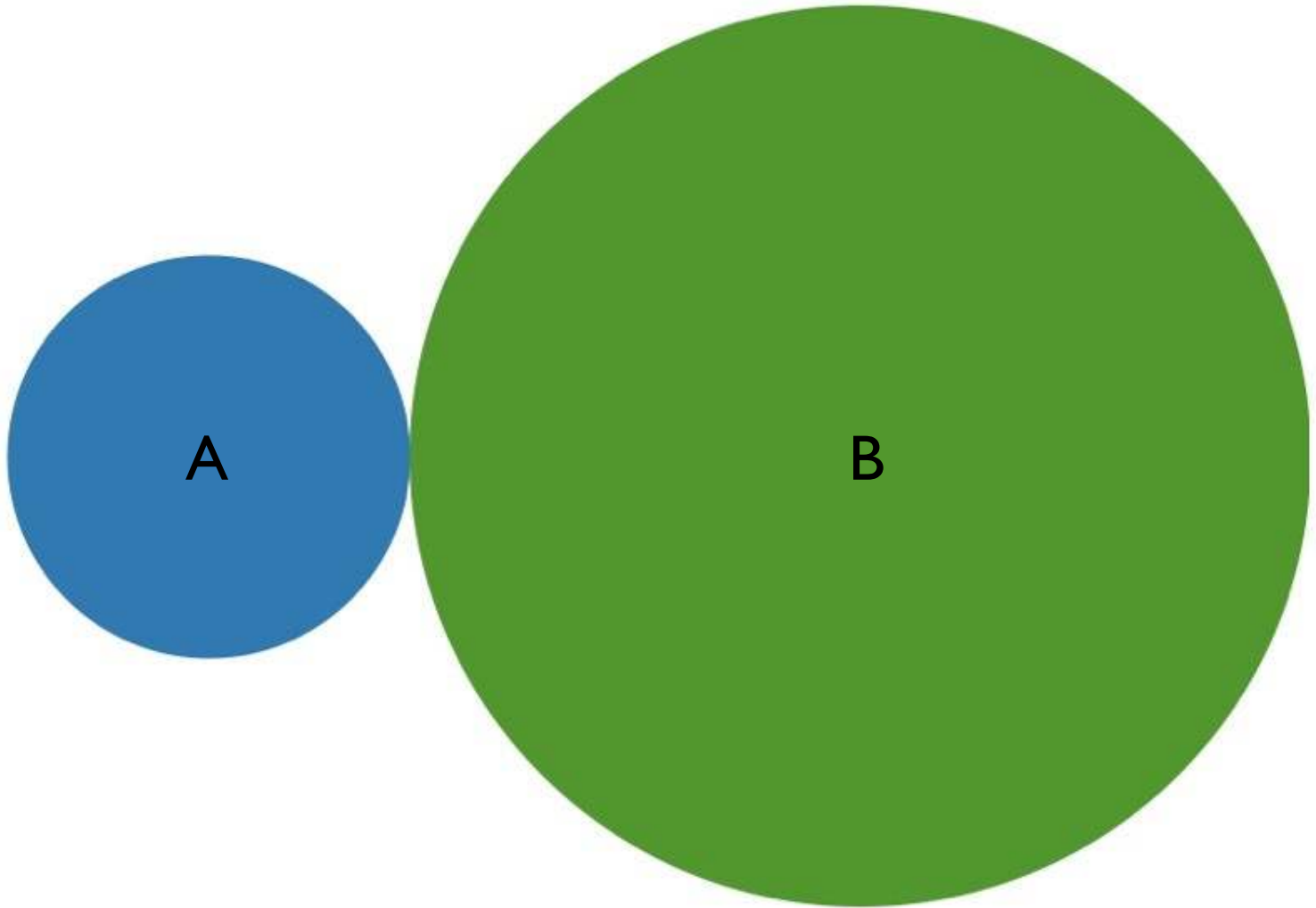


A



B

How much larger area?



How much darker?

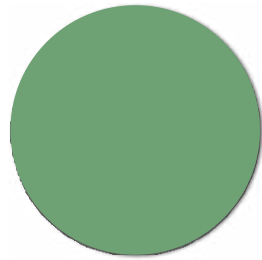


A

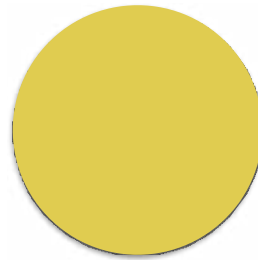


B

How much bigger value?



A



B



Most
Efficient

Position



Length



Slope



Angle



Area



Intensity



Color



Shape



Quantitative

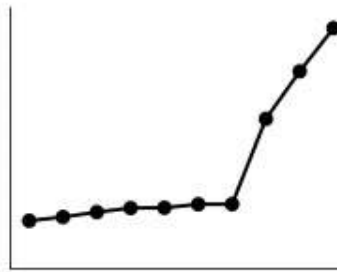
Ordered

Categories

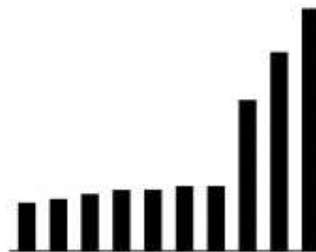
Least
Efficient



Most Effective



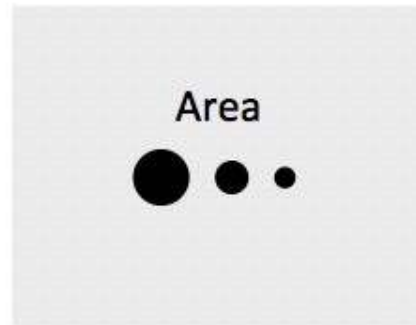
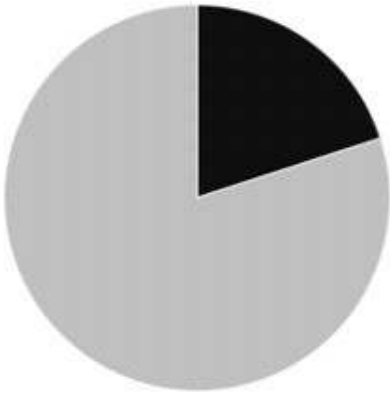
Position



Length

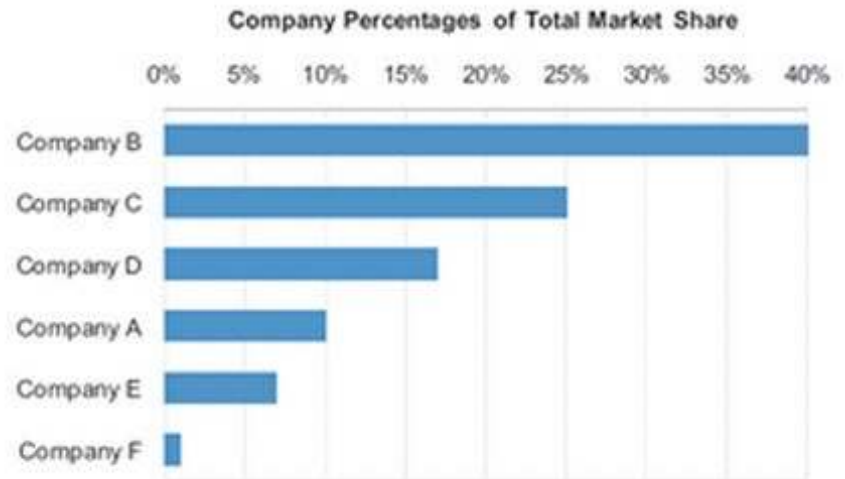
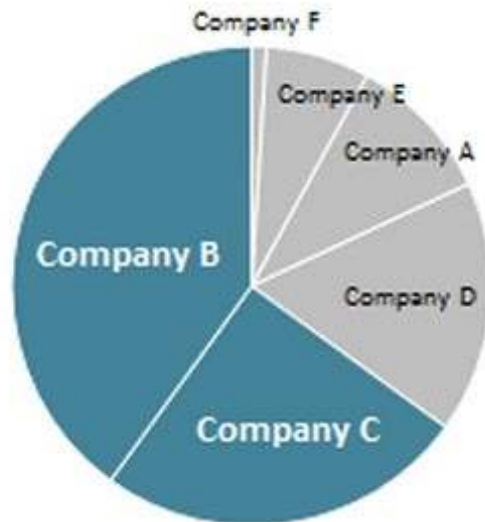


Less Effective



Pie vs. Bar Charts

65% of the market is controlled by companies B and C



Least Effective

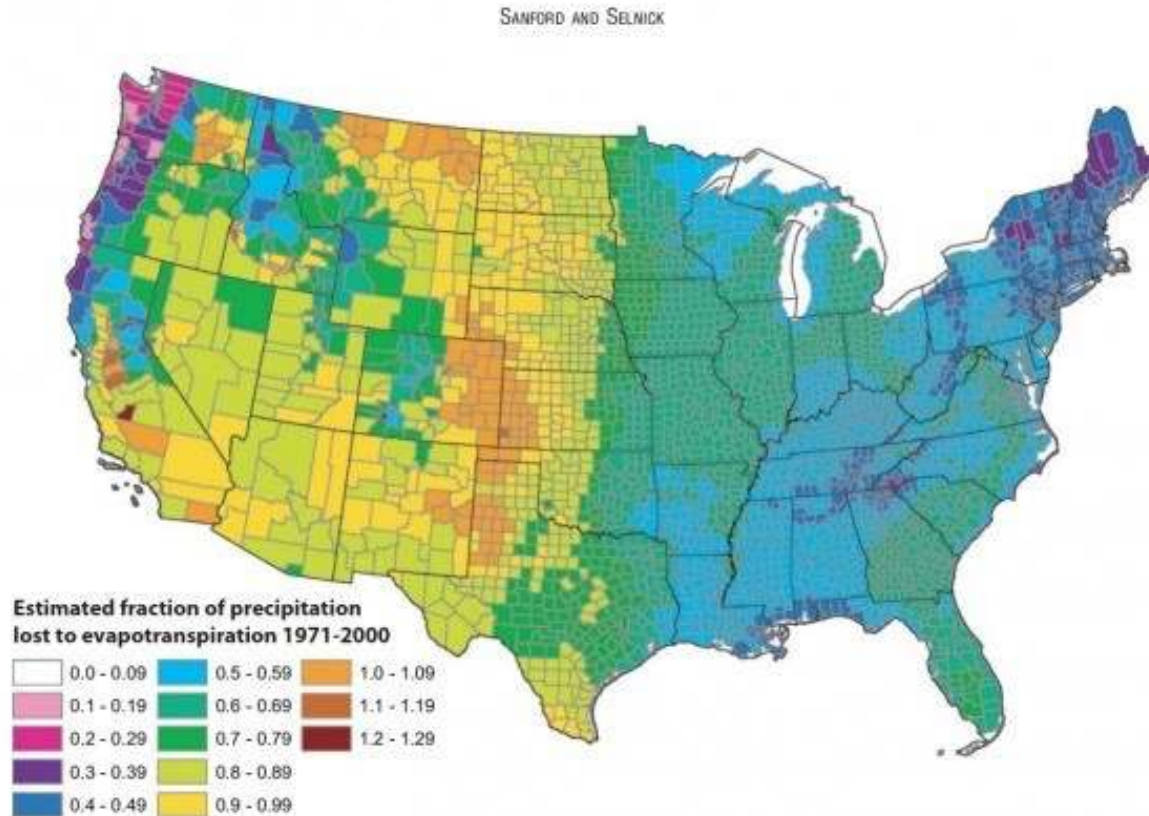
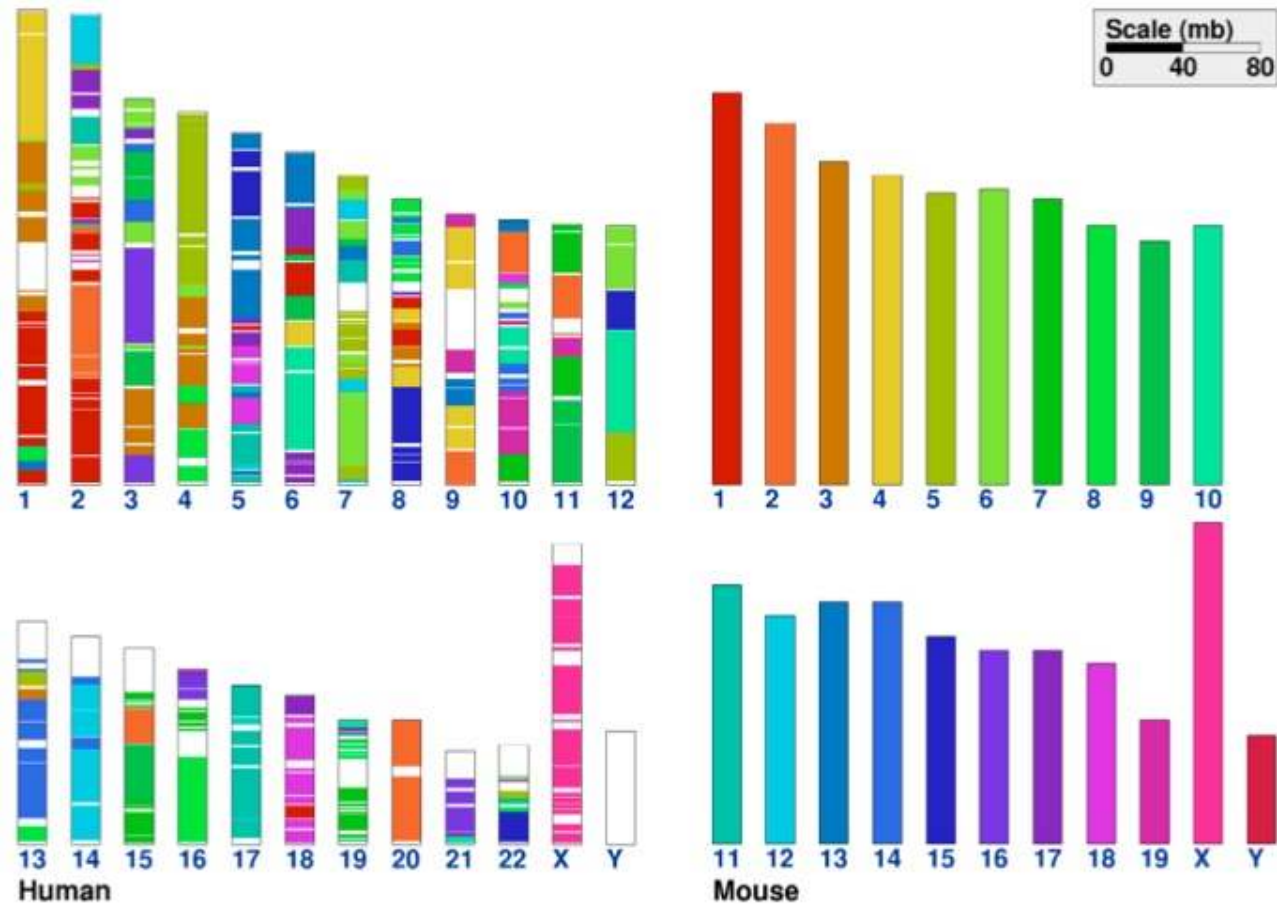


FIGURE 13. Estimated Mean Annual Ratio of Actual Evapotranspiration (ET) to Precipitation (P) for the Conterminous U.S. for the Period 1971-2000. Estimates are based on the regression equation in Table 1 that includes land cover. Calculations of ET/P were made first at the 800-m resolution of the PRISM climate data. The mean values for the counties (shown) were then calculated by averaging the 800-m values within each county. Areas with fractions >1 are agricultural counties that either import surface water or mine deep groundwater.

Use Color Strategically

Color Discriminability



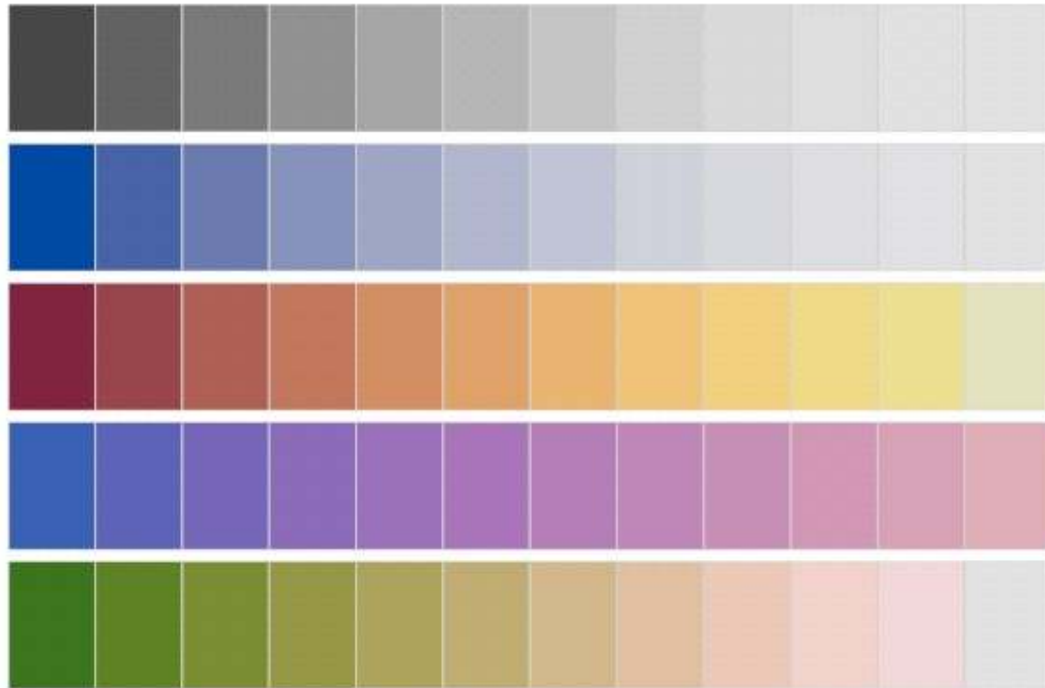
Colors for Categories

Do not use more than 5-8 colors at once



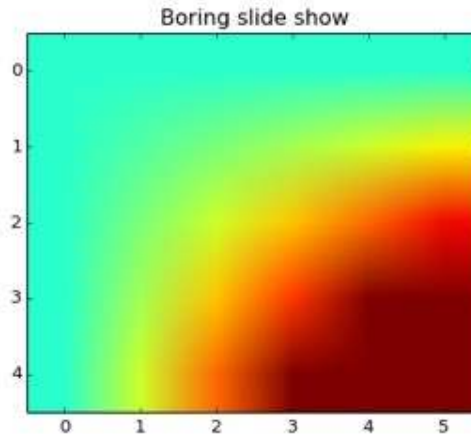
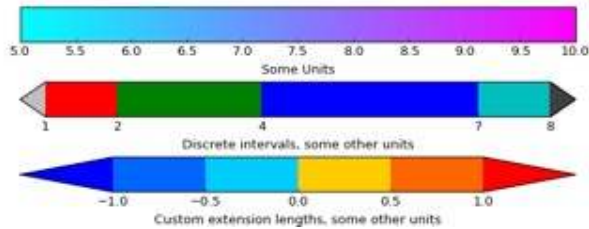
Colors for Ordinal Data

Vary luminance and saturation

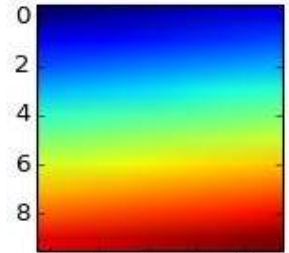


Zeilis et al, 2009, "Escaping RGBland: Selecting
Colors for Statistical Graphics"

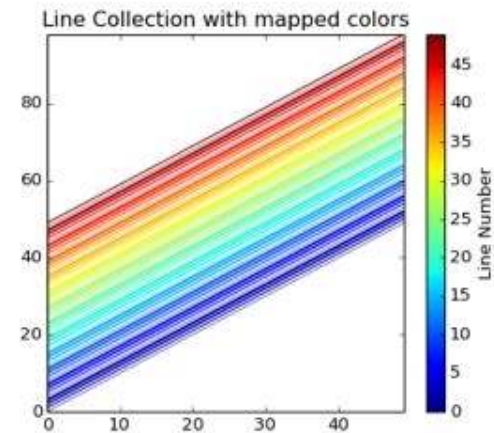
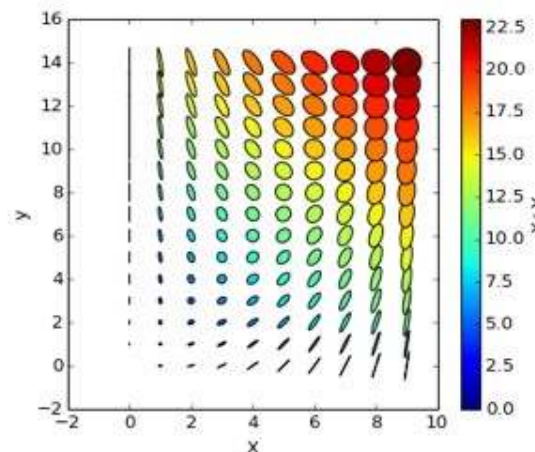
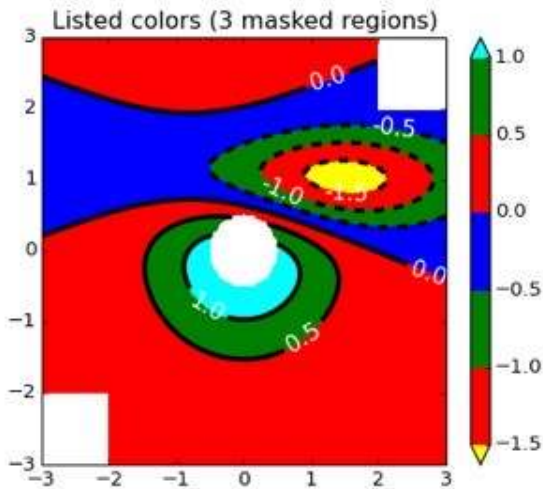
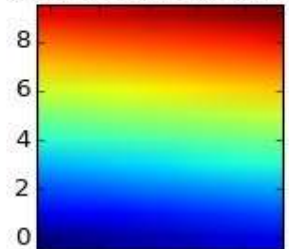
Avoid Rainbow Colors!



blue should be up



blue should be down



Perceptually nonlinear

matplotlib
gallery

Color Blindness



Protanope



Deuteranope



Tritanope

Red / green
deficiencies

Blue / Yellow
deficiency